■ প্রশ্ন কমনের প্রমাণ পত্র ■

প্রমাণপত্র : ২০২৪ সালের পুরাতন সাজেশন্স থেকে কমনকৃত প্রশ্নের হার ৯৮%।

National University

[BBA Hon's 3rd Year; Exam-2024 (Held on- 07/01/2026)]

(Department of Accounting)

Subject Code : 232507

Subject Name : Management Accounting (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question must be written consecutively.]

Part-A

1. Answer any **ten** questions : $1 \times 10 = 10$

(a) Write the name of three types of costs under cost classification for predicting cost behavior.
[অধ্যায়-০২ এর ০৭ নং প্রশ্ন কমন পড়েছে]

(c) What is Product Lifecycle Management (PLM)? *[অধ্যায়-০৩ এর ০৮ নং প্রশ্ন কমন পড়েছে]*

(d) In which case, there in no difference in net income under variable costing and absorption costing?
[অধ্যায়-০১ এর ০৯ নং প্রশ্ন কমন পড়েছে]

(e) What is angle of incidence? *[অধ্যায়-০৩ এর ০৭ নং প্রশ্ন কমন পড়েছে]*

(f) Wht is relevant cost? *[অধ্যায়-০৬ এর ০১ নং প্রশ্ন কমন পড়েছে]*

(g) What is participatory budgeting? *[অধ্যায়-০৭ এর ০৬ নং প্রশ্ন কমন পড়েছে]*

(h) A project's NPV at 10% discount rate is zero (0), what will be the IRR? *[অধ্যায়-০৭ এর ১০ নং প্রশ্ন কমন পড়েছে]*

(i) What is activity based costing? *[অধ্যায়-০৪ এর ০১ নং প্রশ্ন কমন পড়েছে]*

(j) What is flexible budget? *[অধ্যায়-০৮ এর ০১ নং প্রশ্ন কমন পড়েছে]*

(k) What is labor efficiency variance? *[অধ্যায়-০৯ এর ১৮ নং প্রশ্ন কমন পড়েছে]*

(l) What is zero based budget? *[অধ্যায়-০৭ এর ২০ নং প্রশ্ন কমন পড়েছে]*

Part-B

[Answer any **five** questions.] Marks : $4 \times 5 = 20$

2. Distinguish between a master budget and a sales forecast. *[অধ্যায়-০৭ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
3. Briefly describe the prospect of management accounting profession in Bangladesh. *[অধ্যায়-০১ এর ০৫ নং প্রশ্ন কমন পড়েছে]*
4. Show the differences between simple costing system and activity based costing system.
[অধ্যায়-০৪ এর ০৩ নং প্রশ্ন কমন পড়েছে]
5. Discuss the methods of segregation of semi variable cost. *[অধ্যায়-০৩ এর ০৭ নং প্রশ্ন কমন পড়েছে]*
6. *[অধ্যায়-০২ এর **Problem-06** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*
7. *[অধ্যায়-০৩ এর **Problem-01** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*
8. *[অধ্যায়-০৫ এর **Problem-05** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*
9. *[অধ্যায়-০৭ এর **Problem-20** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*

Part-C

[Answer any **five** questions.] Marks : $10 \times 5 = 50$

10. (a) "Management Accounting is the best tools for the management to achieve higher profits and efficient operation". Discuss. *[অধ্যায়-০১ এর ১২ নং প্রশ্ন কমন পড়েছে]*
    (b) Discuss the objectives of preparing cost sheet. *[অধ্যায়-০২ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
11. (a) What are the techniques of management accounting? *[অধ্যায়-০১ এর ০৯ নং প্রশ্ন কমন পড়েছে]*
    (b) Distinguish between financial accounting and management accounting. *[অধ্যায়-০১ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
12. *[অধ্যায়-০৩ এর **Problem-06** নং প্রশ্নের (হুবহু) কমন পড়েছে]*
13. *[অধ্যায়-০৫ এর **Problem-11** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*
14. *[অধ্যায়-০৯ এর **Problem-06** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*
15. *[অধ্যায়-০৭ এর **Problem-14** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*
16. *[অধ্যায়-০৮ এর **Problem-10** নং প্রশ্নের (হুবহু) কমন পড়েছে]*
17. *[অধ্যায়-০৫ এর **Problem-19** নং প্রশ্নের (অনুরূপ) কমন পড়েছে]*

National University

[BBA (Hon's) 3rd Year; Exam-2025 (Will be held-2026/27)]

Subject Code : 232507

Subject Name : Management Accounting (In English)

[Baticrom latest Final Suggestions]

হিসাববিজ্ঞান বিভাগের ছাত্রছাত্রীদের জন্য

Chapter 01 Introduction to Management Accounting

(ব্যবস্থাপনা হিসাববিজ্ঞান পরিচিতি)

Part-A : Brief Questions & Solutions :

ooo 01. **What is accounting?**
Ans : Accounting is an information system that identifies record and communicates the economic events of an organization to interested users.

oo 02. **What is the elaboration of IMA?**
Ans : The elaboration of IMA is Institute of Management Accountants.

ooo 03. **What is financial accounting?**
Ans : Financial accounting is the process of recording all the transactions of the business for reporting and analysis.

ooo 04. **What is the full meaning of AICPA?**
Ans : The full meaning of AICPA is American Institute of Certified Public Accountants.

05 **Define management accounting.** 99%
Or, What is management accounting?
[Test Exam : Suharawardy, Edenm Titumir, Nuzrul, Rajendro & Debendra College;N.U. 2020(Acc.); 2019, 2021 (Mgt.); 2017, 2019, 2021 (Finance)]
Ans : Management accounting is the process of identification measurement, accumulation, analysis, preparation, interpretation and communication of information that assists executives in fulfilling organizational objectives.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 06. **Who is treasurer?**
Ans : The high-ranking official who controls the financial matters of an organization is called treasurer.

oo 07. **What is cost accounting?**
Ans : Cost accounting is the branch of accounting dealing with the classification, recording, allocation, summarization and reporting of current and prospective cost.

ooo 08. **Elaborate GAAP.** 99% *[N.U. 2018, 2020 (Mgt.)]*
Or, What is the full meaning of GAAP?
Ans : The full meaning of GAAP is– Generally Accepted Accounting Principles.

ooo 09. **What are the three major deviation of accounting?**
Ans : The three major deviation of accounting are–
(i) Financial Accounting.
(ii) Cost Accounting.
(iii) Management Accounting.

oo 10. **What is management audit?**
Ans : Management audit is a review of determine whether the policies and procedure specified by the top managers have been implemented.

ooo 11. **What is the elaboration of "ICAB"?** 99%
[N. U. 2021 (Acc.)]
Ans. The elaboration of ICAB is "Institute of Chartered Accountants of Bangladesh."

oo 12. **What is management counseling?**
Ans : Management counselling is an area of public accounting that involves financial planning, control and the development of accounting system.

ooo 13. **Who is management accountant?**
Ans : Management accountant is a person who provide business data and analysis to manager within organization to assist in business decision making and control. *[N.U. 2012 (Old)]*

ooo 14. **What is meant by value chain?**
Or, What is value chain? 99%
[N.U. 2020 (Acc.); 2017 (Mgt.); 2021 (Finance)]
Ans : Value chain is the set of business functions that add value to the products or services of an organization.

০০০ 15. **What do you mean by ethics?** *[N.U. 2019 (Acc.)]*
Ans : The standards of conduct by which one's action are judged as right or wrong, honest or dishonest, fair or unfair.

০০০ 16. **What is business ethics?** 99% *[N.U. 2021 (Acc.)]*
Ans. Business ethics refers to the standards of conduct or moral behavior that apply to business practices.

০০ 17. **What is the full meaning of CIM?**
Ans : The full meaning of CIM is Computer Integrated Manufacturing.

০০০ 18. **What is decision making?**
Ans : Decisions making process is the act of selecting one alternative from a set of available alternatives.

০০০ 19. **What is budget?**
Ans : Budget is a quantitative expression of a plan of action and an aid to coordinating and implementing the plan.

০০০ (20) **What is management process?** 99%
Ans : The management process is one of the methods that aids in structuring, investigating, analyzing, decision-making and communicating business issues. *[N.U. 2023 (Acc.)]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০০ 21. **What is performance report?**
Ans : A performance report is a feedback provided by comparing result with plans and by highlighting variance.

০০ 22. **What do you mean by product life-cycle?**
Or, What is product life cycle?
[N.U. 2020 (Finance)]
Ans : The product life cycle is defined as the period that start with the initial product design and ends with the withdrawl of the product from the market place.

23. What is Product Lifecycle Management (PLM)? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Acc.)]
Ans : Product life cycle management is the succession of strategies used by business management as product goes through its life cycle.

24. What is angle of incidence? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Acc.)]
Ans : In graphical representation of cost-volume-profit analysis the angle which is formed in the right side by the intersection of sales line with the total cost line is known as the angle of incidence.

০০০ 25. **What is meant by benchmarking?**
Ans : Benchmarking is the study of organizations that are among the best in the world at performing a particular task.

০০ 26. **What are the stages of product life cycle?**
Ans : Product life cycles are:
(i) Introducing stage.
(ii) Growth stage.
(iii) Maturity stage.
(iv) Decline stage.

০০০ (27) **Elaborate : ICMAB.** 99%
[N.U. 2016, 2019 (Acc.); 2017, 2019, 2020 (Mgt.); 2019 (Finance)]
Or, What is the elaboration of ICMAB?
Ans : The elaboration of ICMAB is Institute of Cost and Management Accountants of Bangladesh.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০ 28. **What is meant by cost-benefit balance?**
Ans : Cost-benefit balance involves weighing estimated costs against probable benefits, and it is the primary consideration when choosing among accounting system and methods.

০০০ 29. **What is management consulting?** 99%
[N.U. 2018 (Acc.)]
Ans. Management consulting is professional acts, that given necessary guidelines to the managers to achieve organizational goals.

০০০ 30. **Is sunk cost relevant for decision-making?**
[N.U. 2019 (Finance)]
Ans. Sunk cost is not relevant for management decision-making because, it is a historical cost that does not differ between alternatives.

Part- B & C (Short & Broad Questions) :

০০ 01. What is meant by management accounting?
[Ans : হিসাববিজ্ঞান ইন্ট্রোডাক্টরি (৩য় খণ্ড) বইয়ের 4 Page 1.01 No.]
Or, Define management accounting. *[Test Exam : Eden & Edward College; N. U. (Finance)-2016, 2017]*

০০০ 02. **Define management by exception. (MBE).** 99% **[Ans : 4 Page 1.02 No.]**
[Test Exam : Debendra College; N.U. 2013; Accounting-2016; (Finance)-2018]
Or, What do you mean Management by exception?

০০ 03. Why do students of management need to study management accounting? **[Ans : 8 Page 1.12 No.]**
[N.U.BBA 2013; (Mgt.)-2018]

✪✪✪ 04. **(a) Describe the role of management accounting in the decision making process in a business organization. 99% [Ans : 12 Page 1.24 No.]**

[Test Exam : Eden, Jessore city, carmichael, Rajshahi College; N.U. BBA 2005, 2008, 2009, 2014; (Acc)-2018; (Mgt.)-2016, 2018; (Finance)-2018; D.U. BBA (Acc)-2017; (Mgt.) 2016, 2017; (Finance)-2016]

(b) Discuss the career opportunities in management accounting. **[Ans : 10 Page 1.20 No.]**

[N. U. (Acc)-2018; D.U. (Finance)-2016]

✪✪✪ 05. (a) What is meant by product life-cycle management? State the stage of product life-cycle. **99%**

[Ans : 14 Page 1.30 No.] *[Test Exam : Dhaka Eden College, N.U.BBA 2014, Acc-2016; (Mgt.)-2018]*

Or, What is meant by product life cycle management? Discuss the stage of product life cycle.

(b) Define value chain. Name the six primary business functions that make up the value chain. **99%**

[Ans : 15 Page 1.31 No.] *[N.U. BBA- (Acc)-2016]*

✪✪ 06. Discuss the objectives of management accounting. **[Ans : Page No 5; Ques. No 1.04]** *[N. U. (Acc)- 2020]*

✪✪✪ (07) **(a) Discuss the importance of ethics in business. 99% [Ans : Page No 7; Ques. No 1.09]**

[N. U. (Acc)- 2019, 2020]

(b) Discuss the limitations of management accounting. 99% [Ans : Page No 6; Ques. No 1.06]

[N. U. (Acc)- 2020]

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪ 08. (a) Discuss the functions of management accounting. **[Ans : 11 Page 1.22 No.]** *[N. U. (Acc)- 2021]*

(b) Show the differences between management accounting and cost accounting. [Ans : 9 Page 1.15 No.]

Or, Write down the differences between cost accounting and management accounting.

[N. U. (Finance)-2017, 2021; (Acc)-2021]

✪✪✪ (09) **(a) How does management accounting assist the management of a business concern? 99%**

[Ans : 8 Page 1.14 No.] *[N.U (Acc)-2015, 2022]*

(b) Discuss the importance of ethics in management accounting. 99% [Ans : 6 Page 1.07 No.]

[N.U. (Finance)-2018; (Acc)-2017, 2022]

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪ 10. **What factors will you consider at the time of "Make or Buy" decision? [Ans : 14 Page 1.29 No.]**

[N.U. BBA -2012; (Acc)-2016, 2019, 2022; D.U. (Acc)-2016, 2017; (Mgt.) 2016, 2017, 2019]

✪✪✪ 11. **"Management accounting has a broader coverage which includes both internal and external considerations."–Explain. 99% [Ans : 13 Page 1.27 No.]** *[N. U. 2023 (Acc)]*

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

3. Briefly describe the prospect of management accounting profession in Bangladesh.

[Ans : Page No. 634 Question No. 3] *[N. U. 2024 (Acc.)]*

10. (a) "Management Accounting is the best tools for the management to achieve higher profits and efficient operation". Discuss. **[Ans : Page No. 638; Question No. 10 (a)]** *[N. U. 2024 (Acc.)]*

11. (a) What are the techniques of management accounting? **[Ans : Page No. 7; Ques. No. 1.10]**

[N. U. 2010; (Finance)-2018; (Mgt.)2016; (Acc)-2017, 2019, 2021, 2024; (Finance)-2018]

Or, Discuss the various techniques of management accounting.

(b) Distinguish between financial accounting and management accounting. **[Ans : Page No. 8; Ques. No. 1.13]**

Or, Show the differences between financial accounting and management accounting.

[Test Exam : Eden, Dhaka Titumir, Pangsa, satkira, Edward. B.m. & Debendra College; N.U.BBA 2012; (Acc)-2016, 2019, 2024; (Mgt.)-2017, 2018]

Or, What are the differences between management accounting and financial accounting?

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 04, 07, 09 & 11 No. Questions in are most important for the Examinee of 2025 ..... Editor***

Chapter 02 Cost Concepts and Classifications

(ব্যয়ের ধারণা এবং শ্রেণিবিন্যাস)

Part-A : Brief Questions & Solutions :

01 **Define cost and expense.** 99%

[N.U. 2017, 2022(Acc.)]

Ans. Cost : Cost is a forgoing measured in monetary terms that is incurred or potentially incurred to achieve a specific objective.

Expenses : Expenses is a part of cost which is expired and deductible from revenue.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ **02. What is prime cost?**

Or, Define the prime cost.

[Test Exam : Eden, M.M; Edword College; N.U. 2012(Old); 2020 (Acc.); 2016, 2018 (Mgt.); 2021 (Finance)]

Ans : Prime cost is the combined form of direct materials, direct labour and direct expenses.

03 **Mention the two elements of cost.** 99%

[N.U. 2016, 2017, 2020, 2021 (Acc.); 2019, 2020 (Mgt.); 2018 (Finance)]

Or, What are the elements of prime cost?

Or, What are the elements of cost?

Ans : The elements of cost or prime cost are: a. Material, b. Labour.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ **04. Define differential cost.**

[N.U. 2020 (Acc.); 2017 (Finance); 2021 (Mgt.)]

Or, What is differential cost?

Ans : A difference in cost between two alternatives is called differential cost.

○○○ **05. What do you mean by administrative overhead?**

Ans : All executive, organizational and clerical costs associated with the general management of an organization is called administrative overhead.

○○○ **06. What is opportunity cost?** 99%

[N.U. 2013, 2015 (Old); 2019, 2022 (Acc.); 2016, 2020, 2022 (Mgt.); 2016, 2020 (Finance)]

Ans : The potential benefit that is lost or sacrificed when the selection of one course of action makes it necessary to give up a competing course of action is called opportunity cost.

○○○ **07. What is incremental cost?** *[N.U. 2019 (Finance)]*

Ans : An increase in cost between any two alternatives is called incremental cost.

○○○ **08. What is sunk cost?** 99%

[N.U. 2012, 2014, 2015(Old); 2017, 2019 (Acc.); 2016, 2018, 2021 (Mgt.); 2021 (Finance)]

Ans : Sunk cost is the cost that has already been incurred and that cannot be changed by any decision made now or in the future.

○○ **09. What is the marketing and selling overhead?**

Ans : Sales and marketing overheads are costs incurred in the marketing of a companies products or services to potential customers.

○○○ **10. What is cost center?** *[N.U. 2021 (Acc.)]*

Ans : Cost center is a segment of a company that has a control over generating revenue or the use of investment of funds.

○○○ **11. What is cost behavior?**

[N.U. 2015(Old); 2017 (Mgt.)]

Or, What is meant by cost behavior?

Ans : The way which a cost will react or respond to changes in the level of business activity is called cost behaviour.

12. Write the name of three types of costs under cost classification for predicting cost behavior. *[২০২৪ সালে যা এসেছিল]* *[N.U. 2024 (Acc.)]*

Ans : The name of three types of costs under cost classification for predicting cost behavior are variable cost, fixed cost and semi-variable cost.

○○○ **13. What is the direct costing?**

Ans : A cost that can be obviously and physically traced to a unit of product or other organization segment is called direct costing.

○○○ **14. What is the periodic cost?**

Ans : All costs that are matched against revenue on a time period basis; such costs consist of selling (marketing) and administrative expenses is called periodic cost. *[N.U. 2020 (Finance)]*

○○○ **15. What is product cost?** 99%

[N.U. 2020 (Acc.); 2019 (Finance); 2021 (Mgt.)]

Ans : All costs that are involved in the purchase or manufacture of goods are called product costs.

16 **What is conversion cost?** 99%

[N.U. 2012, 2014 (Old); 2016 2019 (Acc.); 2018 (Mgt.)]

Ans : Conversion cost is the combined form of direct labour and manufacturing overhead.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ **17. What is indirect cost?**

Ans : The cost that cannot be easily and conveniently traced to the particular cost object is called indirect cost.

○○ **18. What is cost of goods manufactured?**

Ans : The materials, labor and overhead costs that have gone into the products produced during a period are called the cost of goods manufactured.

০০০ 19. **What is finished goods?**
Ans : Goods that are complicated as to manufacturing but not yet sold to customers is called finished goods.

০০০ 20. **What is manufacturing overhead?** *[N.U. 2015 (Old)]*
Ans : All costs associated with the manufacturing process except direct materials and direct labour are called manufacturing overhead.

০০০ 21. **What is direct materials?**
Ans : The materials that become an integral part of a finished product and can be directly traced to it are called direct materials.

০০০ 22. **What is direct labor?**
Ans : Those factory labor costs that can be easily traced to individual units of product are called direct labour costs.

০০০ 23. **What is cost sheet?** 99% *[N.U. 2022(Acc.); 2015 (Old); 2018 (Mgt.)]*
Or, What is cost statement?
Ans : Cost sheet is a statement which is prepared usually to present the detailed costs of total output during a period at sequence.

০০০ 24. **What is the indirect labor?**
Ans : The labour cost that cannot be conveniently traced directly to particular products is called indirect labour.

০০ 25. **What is work in process?** *[N.U. 2022(Acc.)]*
Or, What do you meant by work-in-progress?
Ans : Work in process consists of units of product that are only partially complete and will require further work before they are ready for sale to the customers.

০০০ 26. **Give two examples of static cost.** 99%
Ans : Two examples of static cost are (i) Monthly office rent & (ii) Salaries of permanent staff. *[N.U. 2023 (Acc.)]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০ 27. **What is the indirect material?**
Ans : Indirect materials are those materials which cannot be conveniently identified with individual cost units.

০০০ 28. **What is shutdown cost?** 99% *[N.U. 2018 (Acc.)]*
Ans. A cost that will still be required to be incurred even though a plant is closed or shutdown for a temporary period is known as shutdown cost.

০০০ 29. **In which case, more net profit arise in variable costing?** *[N.U. 2017 (Mgt.)]*
Ans. More net profit arises under variable costing when there is a decrease in inventory levels because only variable manufacturing costs are included in the cost of the product, so fixed manufacturing overhead is treated as a period cost and expenses in full during the period.

০০০ 30. **What is target costing?** 99% *[N. U. 2023 (Acc.)]*
Ans : Target costing implies developing product cost in terms of what the market will pay for a product with specific characteristics.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part-B & C (Short & Broad Questions) :

০০০ 01. (a) Discuss the cost concept. **[Ans : রকেট হিসাববিজ্ঞান ইজি প্লাস (৩য় খণ্ড) বইয়ের 21 Page 2.18 No.]**
Or, Define cost concept.
(b) Explain the different elements of cost. **[Ans : 23 Page 2.22 No.]** *[N.U (Acc)-2015]*

০০০ 02. **"All future costs are relevant"-Do you agree? Why?** 99% **[Ans : 27 Page 2.35 No.]**
[N.U. (Acc)-2015; (Mgt.)-2017; D.U. (Mgt.)-2016]
Or, "All future costs are relevant in decision making"-Do you agree? Explain.

০০ 03. Briefly describe the elements of cost of a manufacturing concern. **[Ans : 22 Page 2.19 No.]**
[Test Exam- Anandamohan, Thakurgan, Tongi, Pangsha, Rajendra, Rajshahi College; N.U (Acc)-2015]

০০০ 04. Discuss the cost classification for decision making. **[Ans : 23 Page 2.24 No.]** *[N.U (Acc)-2016]*

০০ 05. Distinguish between differential cost and marginal cost. **[Ans : 25 Page 2.28 No.]** *[N.U (Acc)-2014]*

০০ 06. Show the difference between fixed and variable cost. **[Ans : 19 Page 2.09 No.]**
[Test Exam : Eden, Debendra, Rafiqual, Islam mohila, Rajshahi, chadaga College; N.U. (Acc)-2019]

০০০ 07. **Show the difference between product cost and period cost.** 99% **[Ans : 21 Page 2.17 No.]**
Or, Discuss the differences between product cost and period cost. *[N. U. (Acc)-2019, 2021]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০ 08. Describe the various type of cost according to cost behaviour. **[Ans : 26 Page 2.30 No.]**
Or, Classify cost on the basis of significance and behaviour. *[N. U. BBA-2013, (Acc)-2021; (Mgt.)-2018]*

ooo 09. **"Variable cost is fixed and fixed cost is variable" Explain. 99% [Ans : 27 Page 2.36 No.]**
[Test Exam : Edward, Bhola, Rajbri, M.M, Chittagong city College; N.U. 2007; 2022(Acc.)]

Or, "Fixed cost per unit is variable but variable cost per unit remains fixed" –Explain

ooo (10) **"If a product line is generating a loss, then it should be discontinued."– Do you agree? Explain. 99%**
[Ans : 25 Page 2.27 No.] *[N.U. 2023 (Acc.)]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

10. (b) Discuss the objectives of preparing cost sheet. **[Ans : Page No. 638; Question No. 10 (b)]** *[N.U. 2024 (Acc.)]*

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 02, 07 & 10 No. Questions in are most important for the Examinee of 2025 ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

ooo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2014 No–03; (Finance) 2016 No–05; (Mgt) 2021 No–4; MBA Preli (Mgt.) 2018 No–06**

As a cost accountant of Evan Fabrics/Meena Ltd /From the following relevant information, prepare a cost sheet:

Direct materials	Tk. 1,00,000
Direct labour	80% of material
Factory on cost	50% of direct labour
Office on cost	75% of factory on cost
Selling on cost	50% of office on cost
Profit	25% of sales

Required : (1) Prime cost ; (2) Factory cost ; (3) Cost of production ; (4) Cost of sales ; (5) Sales ; (6) Percentage of total cost of sales on sales.

Solution : ব্যাংকার্স হিসাববিজ্ঞান ইন্টিগ্রেটস (৩য় খণ্ড) বইয়ের ৩৩ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 05**

Amazon Company produces radios for cars. The following cost information is available for the period ended December 31, 2019 :

Materials put into production Tk. 1,20,000, of which Tk. 80,000 was for direct materials. Factory labour costs for the period Tk. 90,000, of which Tk. 25,000 was for indirect labour. Factory overhead costs for utilities Tk. 40,000. Work-in-process beginning and ending value were Tk. 12,000 and Tk. 20,000 respectively.

Required : Calculate cost of goods manufactured (show prime costs and total manufacturing costs in your calculations).

Solution : ৩৪ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2019 No– 06; (Finance) 2021 No–08**

The following data are related to the Akter Company Ltd./Adams Company Ltd. :

	Inventories	
	Ending (Taka)	**Beginning (Taka)**
Finished goods	95,000	1,10,000
Work-in-process	60,000	50,000
Direct materials	85,000	80,000

Cost incurred during the period :	Taka
Cost of goods available for sales	6,84,000
Total manufacturing cost	5,84,000
Factory overhead	1,67,000
Direct materials used	1,90,000

Required :

(i) Direct labour cost.

(ii) Direct materials purchased during the period.

Solution : ৩৪ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2017 No–02**

The following data related to Depu Limited :-

Account titles	Taka	Taka
Inventories	**Opening**	**Ending**
Material	9,000	12,000
Work-in-process	6,000	5,000
Finished goods	5,700	11,600

Direct labour Tk. 7,500, Administrative expense Tk. 3,000, Material purchase Tk. 40,000, Factory expense Tk. 4,000.

Required : Cost of goods sold.

Solution : ৩৭ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd 2013 No–10; (Acc.) 2020 No– 11; 3rd (Finance) 2016 No–11; 2020 No– 13**

The Surma Company/The Apex Company produce different types of leather goods. The sales revenue for the year Tk. 12,00,000. The cost related different information's during the year were as follows:

Account Titles	Taka
Raw material purchased	50,000
Freight in	2,000
Opening work-in-process	80,000
Opening finished goods	40,000
Opening raw materials	2,50,000
Closing work-in-process	70,000
Closing finished goods	60,000
Closing raw materials	60,000
Productive wages	1,70,000
Chargeable expense	50,000
Factory overhead	2,50, 000
Selling expense	40,000
General and administrative expense	10,000
Sale of scrap	5,000

Prepare a statement showing the following :

(i) Raw material used;

(ii) Prime cost;

(iii) Cost of manufacturing;

(iv) Cost of goods manufactured;

(v) Cost of goods available for use;

(vi) Cost of goods sold;

(vii) Cost of sales;

(viii) Profit or loss.

Solution : ৪১ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2017 No–10(b)**

The cost of production related information of Rahima Ltd. are given below :

Account Titles	01-01-2014	31-12-2014
Raw materials	10,000	12,000
Work-in-rocess	20,000	15,000
Finished goods (120 units)	30,000	
Additional information :		Taka
Purchase of raw material		1,60,000
Direct wages		1,20,000
Factory overhead		85,500
Sales (1,500 units including opening inventory)		4,50,000

Closing finished goods 80 units, administrative expense 20% of factory overhead and selling expense 5% on sales. You are required to calculate units of goods produced and value of ending inventory. Draw up a statement of cost showing the COGS and profit.

Solution : ৪৩ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊗⊗ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–6**

The following figures for the year ended December 31, 2022 were extracted from the records of a manufacturing company :

Inventory of finished goods (1-1-2022) (5,000 units)	45,000
Raw materials purchase	2,60,000
Direct wages	1,00,000
Sales (45,000 units)	7,00,000

Ending finished goods (10,000 units). Factory overhead 60% of direct wages. Administrative overhead Tk. 50,000, selling overhead Tk. 2 per unit.

Prepare a cost sheet for the year ended December 31, 2022.

Solution : ৬৩৫ পৃষ্ঠার **Question No-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Management 2018 No–11; (MBA Preli.) (Management) 2017 No–2(c)**

The following are the information from records of Haque and Company/ Star and Company for six months ended on 30th June, 2018 :

Account Titles	Taka
Raw materials	18,000
Direct labour	11,000
Indirect labour	5,000
Factory supplies	1,000
Salaries	9,000
Others factory expense	2,500
Miscellaneous selling expense	2,000
Administrative expenses	1,500

Depreciation (75% relates to manufacturing, 15% administrative and 10% selling and distribution) value of assets as on 1st January, 2018 was Tk. 4,00,000 @ 5% per annum is to provided $\frac{2}{3}$ of salary is for office and the balance of sales. Total 5,000 units produced and sold during the six months, Selling price per unit is Tk. 16.

Prepare a statement showing the total cost of sales and profit earned during the year

Solution : ৪৮ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2022 No– 13**

Asia Ltd. sells 1,200 units of a product at Tk. 92 per unit. Cost of direct materials is 30% of the cost of goods sold. The ratio of direct wages and factory overhead is 5 : 2. The administrative and selling expenses amount to Tk. 21,200. The net profit during the current year is Tk. 24,400.

The company plans to sell 1,500 units next year at Tk. 96 per unit. Wages rates and raw materials prices are expected to increase by 20% and 10% respectively. Factory overhead is applied on the basis of direct labour cost. Other cost will remain unchanged. Show the difference in net profit of the two years.

Solution : ৪৯ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

oo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2016 No–5**

In 2016 Tanzim Company had sales of Tk. 1,99,50,000 with Tk. 1,15,71,000 variable and Tk. 76,23,000 fixed costs. 2017 sales are expected to decrease 15% and the cost relationship is expected to remain constant. (The fixed costs with not change) Determine Tanzim Company's operating income or loss for 2017.

Solution : ৫০ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–6**

The following data relates to the Horizon Manufacturing Company of a standard product during four weeks period of June 30, 2024 :

Account titles	Tk.
Raw material used	Tk. 5,000
Wages	Tk. 7,000
Machine hours worked 2,000 hours	
Machine hour rate	Tk. 1.00
Office overhead : 25% on work cost	
Selling overhead per units	Tk. 0.10
Units produced :	25,000 units
Units sold :	2,000 units @ Tk. 1 per unit

Required : (i) Cost sheet; (ii) Income statement.

Solution : ৫২ পৃষ্ঠার **Problem-29** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Management) 2016 No–11**

The cost department of the Swagata Corporation made the following data and costs available for the year 2016 :

Account Titles	Inventories	
	January 1	December 31
Raw materials	34,200	49,300
Work-in-process	81,500	42,350
Finished goods	48,600	?
Depreciation on factory equipment	21,350	
Interest earned		6,300
Raw materials purchased		3,64,000
Direct labour		1,62,500
Indirect labour		83,400
Freight in		8,600
Miscellaneous factory ovethead		47,900
Purchased discounts		5,200

Finished goods inventory : January 1 : 300 Units ; December 31 : 420 units, all from current years production.
Sold during 2016 : 3,880 units @ Tk. 220 per unit.

Required :

(i) Statement of cost of goods sold:

(ii) Income statement.

Solution : ৫৬ পৃষ্ঠার **Problem-33** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Management) 2020 No– 11**

The books of Walton Manufacturing Company engaged in assembling refrigerators, show the following information for the six months ended June 30, 2021.

Raw materials purchase	20,82,500
Inventory June 1, 2021	
Raw materials	2,50,000
Finished goods (50 refrigerators)	41,200
Direct labour	20,50,000
Factory overhead	8,16,300
Selling expense	4,28,000
General and administrative expense	4,96,000
Sales (6,000 refrigerators)	63,60,000
Inventory June 30, 2021	
Raw materials	1,50,000
Finished goods 250 refrigerators	
No unfinished work on hand.	

You are required to calculate :

(a) A statement cost of goods manufactured.

(b) The number of units manufactured.

(c) The unit of refrigerators manufactured.

(d) The profit per unit sold.

Solution : ৫৭ পৃষ্ঠার **Problem-34** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old)-2014 No–10; (Finance) 2018 No–13**

A manufacturing company produces bicycles. The company had 260 units of opening inventory costing Tk. 700 per unit in 2017.

During the year 2014 the company sold 2,150 bicycles at a gross profit of 20% on sale. cost of material per unit was Tk. 420 Direct labour cost per bicycle was 40 hours at Tk. 4 per hour. Factory overhead was charged at Tk. 5 per direct labour hour. Administrative and selling expense amounted to Tk. 1,20,000. The finished inventory at close was 110 units.

Prepare :

(i) Statements of cost of goods sold;

(ii) Income statement.

Solution : ৫৯ পৃষ্ঠার **Problem-36** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–16**

A factory produces and supplies goods according to customers' specification. From the following particulars for the year 2020 prepare a statement of cost :

Inventory of raw materials (January 1, 2020)	Tk. 25,000
Inventory of finished goods (January 1, 2020)	Tk. 50,000
Inventory of raw materials (December 31, 2020)	Tk. 37,500
Inventory of finished goods (December 31, 2020)	Tk. 30,000

Purchase of raw materials	Tk. 1,62,500
Direct wages	Tk. 1,00,000
Factory overhead	Tk. 62,500
Office and administrative overhead	Tk. 31,250
Sales	Tk. 4,00,000

The factory is about to send a tender for supply of machine. The costing department estimates that the materials required would cost Tk. 60,000 and direct wages would cost Tk. 40,000. Factory overhead is charged at a percentage on prime cost and office and administrative overhead is charged at a percentage on factory overhead.

Compute the price of the tender to earn a profit of 20% on total cost.

Solution : ৬৪ পৃষ্ঠার **Problem-39** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Management) 2017 No– 11; D.U. BBA (Hon's) 3rd (Acc.) 2020 No– 2(b)**

The following figures are available from the books of M/S Sandeep Enterprise for the year ended 31st December. 2015 :

Accounts titles	Taka
Materials stock on 1-1-2015	1,000
Materials stock on 31-12-2015	2,000
Materials purchased during 2015	10,000
Wages	7,500
Factory overhead	4,500
Administrative overhead	4,200
Selling overhead	5,250
Profit for the year	6,090

In the year 2016 the factor, receives an order for a job which will require materials Tk. 1,200 and wages Tk. 750. The management of the enterprise intends to earn a profit 10% higher than the percentage of profit earned in 2015 and assume that the factory overhead has gone up by 15% and selling and administrative overhead has gone down by 20% in 2016. Further assume that factory overhead is recovered as a percentage of wage and selling and administrative overheads as a percentage of works cost.

Required :

(i) Prepare a cost sheet for 2015;

(ii) Ascertain the sales price of the job order received in 2016.

Solution : ৬৫ পৃষ্ঠার **Problem-40** নং প্রশ্নের উত্তর দেখুন।

Chapter 03 Absorption and Variable Costing

(গ্রসন এবং পরিবর্তনশীল ব্যয়)

Part-A : Brief Questions & Solutions :

ooo 01. **What is variable cost? 99% (পরিবর্তনশীল ব্যয় কী?)** *[N.U. 2020 (Mgt.)]*

Ans : A cost that varies in total, in direct proportion to change in the level of activity is called variable cost.

ooo 02. **Which costing method is used for external reporting?** *[N.U. 2017 (Mgt.)]*

Ans : Absorption costing is used for external reporting.

ooo 03. **Which costing method is also known as 'Full costing'?** *[N.U. 2014 (Old); 2018 (Mgt.)]*

Ans : Absorption costing is also known as full costing.

oo 04. **What is inventory? (মজুদ পণ্য কী?)**

Ans : Inventory is the part of product cost that has not yet been sold and is it shown on the balance sheet as an asset.

oo 05. **What is fixed cost?** *[N.U. 2016 (Mgt.)]*
Ans : The cost which is not affected by change in activity is called fixed costs.

06. What is mixed cost? [২০২৪ সালে যা এসেছিল]
[N.U. 2024 (Acc.)]
Ans : Mixed cost is a cost that contains both fixed and variable components.

oo 07. **Which costing is also known as "Direct costing"?** *[N.U. 2017 (Finance)]*
Ans : Variable costing is also known as direct costing.

08 **Define absorption costing.** 99%
[Test Exam : Dhaka, Eden & Titumir College; N.U. 2019, 2021 (Acc.); 2021 (Finance)]
Or, What is absorption costing?
Or, What do you mean by absorption costing?
Ans : A costing method that includes all manufacturing cost-direct materials, direct labour and both variable and fixed manufacturing overhead in the cost of a unit of product.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 09. **What is inventorial cost?** 99% *[N.U. 2018 (Acc.)]*
Ans : All costs that are involved in the purchase of manufacture of goods. In the case of manufactured goods, these costs consist of direct materials, direct labour and manufacturing overhead costs used in the production process.

oo 10. **What do you mean by cost of goods sold?**
Ans : In the period of sales, the product costs are recognized as an expense called cost of goods sold.

oo 11. **What is throughput?**
Or, What is super variable costing? (অতি পরিবর্তনশীল ব্যয় কী?)
Ans : The total volume of production through a facility during a period is called throughput or super variable costing.

12 **What is segment reporting?** 99% *[N.U. 2023 (Acc.)]*
Ans : An income statement or other report in an organization in which data are divided according to product line, division, territories or similar organizational segments.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo 13. **What is common fixed cost? (সাধারণ স্থায়ী ব্যয় কী?)**
Ans : A cost that cannot be identified with any particular segment of an organization is called common fixed cost. Such cost which are also known as indirect cost.

ooo 14. **What is normalized overhead rate?**
Ans : A rate based on the average activity of many past and present periods rather than only on the expected activity of the current period.

ooo 15. **What is traceable fixed cost? (চিহ্নিত স্থায়ী ব্যয় কী?** 99% *[N.U. 2018 (Acc.)]*
Ans : A cost that can be identified with a particular segments and that arises because of the existence of that segment is called traceable fixed cost.

ooo 16. **What is segment margin?** *[N.U. 2020 (Finance)]*
Ans : The amount remaining from the sales of a segment after it has covered all of its own variable and fixed cost is called segment margin.

oo 17. **What is operating income?**
Ans : Operating income is measure of profitability that tells investors how much revenue will eventually become profit for a company after covering operating expenses.

18 **What do you mean by just-in-time?** 99%
[N.U. 2017 (Acc.); 2017, 2020 (Finance)]
Or, Define just-in-time.
Or, What is JIT?
Ans. JIT is production and inventory control system in which materials are purchased and units are produced only as needed to customer demand.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 19. **What do you mean by business segment?**
Ans : Business segment means each of the parts which is or may be divided price, section part, portion division etc. *[N.U. 2016 (Finance)]*

ooo 20. **Write down the formula of budgeted production volume.** 99% *[N.U. 2017 (Finance)]*
Ans. Budgeted production volume = Budgeted Sales + Required ending inventory – Estimated opening.

oo 21. **What is step fixed cost?**
Ans : A type of expenses that is more or less constant over a low level shift in activity, but which changes incrementally when activity shifts substantially.

22. In which case, there in no difference in net income under variable costing and absorption costing? [২০২৪ সালে যা এসেছিল]
[N.U. 2024 (Acc.)]
Ans : There is no difference in net income under variable costing and absorption costing when there is no change in inventory levels.

Part- B & C (Short & Broad Questions) :

✪✪✪ 01. (a) Definition of variable costing. [Ans হিসাববিজ্ঞান ইজিপ্রেস (৩য় খণ্ড) বইয়ের 71 Page 3.01 No.]

Or, What do you mean by direct costing?

(b) Discuss the feature/characteristics of variable costing. **[Ans : 71 Page 3.02 No.]**

[Test Exam : Dhaka, Eden, Lalmatia, Mohila, Monsar, Mohamad mohsin & Suhrawardy College]

✪✪✪ 02. **Discuss the product cost and period cost.** 99% **[Ans : 74 Page 3.11 No.]** *[N.U. (Acc)-2017]*

✪✪✪ (03) **Show the differences between variable costing and absorption costing.** 99%

[Ans : Page No 75; Ques. No 3.14] ***[N.U. BBA -2009, 2013; (Mgt.)-2016; (Acc)-2015, 2017, 2018, 2020; D.U. (Accounting & Management)-2016]***

Or, Describe the reasons for difference in net income between variable costing and absorption costing.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪ 04. Only variable cost can be differential cost. Do you agree? Explain. **[Ans : 72 Page 3.05 No.]**

[Test Exam- Shadot College]

✪✪✪ 05. **When and why profits will be higher in case of absorption costing and variable costing?** 99% **[Ans : 76 Page 3.15 No.]** *[N.U. 2023 (Acc)]*

✪✪ 06. Difference between variable costing and marginal costing? **[Ans : 74 Page 3.12 No.]**

✪✪✪ 07. **"Value of inventory under full costing is greater than variable costing" –Explain with example.** 99% **[Ans : 75 Page 3.13 No.]** *[N.U. 2023 (Acc)]*

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

5. Discuss the methods of segregation of semi variable cost. **[Ans : Page No. 635; Question No. 5]** *[N.U. 2024 (Acc)]*

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 03, 05 & 07 No. Question in are most important for the Examinee of 2025 ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

✪✪✪ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2012 No– 03; (Finance) 2018 No–06; 2020 No– 09**

Sonali Company Ltd. manufactures bicycle for the use on city street :

Beginning inventory:	
Units produced	10,000
Units sold	8,000
Selling price per unit	Tk. 500
Manufacturing cost :	
Direct materials cost per unit	Tk. 120
Direct labour cost per unit	Tk. 140
Variable manufacturing overhead per unit	Tk. 50
Fixed manufacturing overhead	Tk. 6,00,000

Required :

(i) Determine the unit product cost under absorption costing;

(ii) Determine the unit product cost under variable costing.

Solution : হিসাববিজ্ঞান ইজিপ্রেস (৩য় খণ্ড) বইয়ের ৭৯ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

Problem-02 ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–7**

ACI Ltd. produces a single product and supplies the following cost data :

Production during the year	20,000 units
Sales during the year	15,000 units
Direct material cost	Tk. 1,50,000
Direct labour cost	Tk. 1,00,000
Variable factory overhead	Tk. 60,000
Fixed factory overhead	Tk. 40,000

Required :

(a) Cost per unit and value of ending inventory under variable costing.

(b) Cost per unit and value of ending inventory under absorption costing.

Solution : ৬৩৬ পৃষ্ঠার **Question No-07** নং প্রশ্নের উত্তর দেখুন।

Problem-03 ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–7**

Rangdhonu Ltd. produced 32,000 units of product during the year 2022, 27,000 units were sold @ Tk. 200 each. Cost of this product was :

Material Tk. 7,00,000. Labour (direct) Tk. 5,00,000

Factory OH :	
Fixed	Tk. 1,52,000
Variable	Tk. 1,25,000

Marketing and administrative expenses for the year total Tk. 1,10,000, all are fixed. You are required production cost per unit using variable costing method.

Solution : ৭৮ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Problem-04 ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2016 No–06**

NUHA Ltd. has the following data :

Account Titles	Year 1 Tk.	Year 2 Tk.
Beginning inventory as per variable costing	Nill	40,000
Beginning inventory as per absorption costing	Nill	58,000
Ending inventory as per variable costing	40,000	Nill
Ending inventory as per absorption costing	58,000	Nill
Profit as per variable costing	50,000	50,000
Profit as per absorption costing	68,000	32,000

Prepare a reconciliation statement.

Solution : ৮২ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

Problem-05 ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–9**

Ratul Enterprise supplied you the following information about its production and sales :

Selling price Tk. 16 per unit

Variable cost Tk. 8 per unit

Production is 10,000 units

Beginning inventory was 8,000 units

Ending inventory is 6,000 units

Variable selling and administration expense Tk. 0.5 per unit

Fixed selling and administration expense Tk. 10,000

Fixed factory overhead was Tk. 40,000 for 20,000 units.

You are required to :

(i) Find net income under absorption costing

(ii) Explain the reason for the difference in net income between absorption costing and direct costing method.

Solution : ৮৩ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2017 No–04**

Dhaka Manufacturing Company produces one product. Its cost includes direect materials Tk. 10 per unit, direct labour Tk. 8 per unit, variable manufacturing overhead Tk. 6 per unit, fixed manufacturing overhead Tk. 2,50,000 and selling and administrative overhead Tk. 30,000. In October 2012, Dhaka Manufacturing Co. Produced 25,000 units and sold 20,000 units at Tk. 50 each.

Required :

(i) Calculate product cost per unit under variable costing.

(ii) Prepare a variable costing income statement.

Solution : ৮৪ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2016 No–11**

The following particulars are available from the books of a company for the year ended 2016 :—

Sales : 1,50,000 kg; sales price Tk. 20 per kg.

Finished goods inventory (1-1-2016) 24,000 kg.

Finished goods inventory (31-12-2016) 35,000 kg.

Manufacturing cost:

Variable cost per kg, Tk. 8.

Fixed factory overhead Tk. 1,60,000 (normal capacity 1,60,000 kg).

Marketing and Administrative expenses :

Variable cost per kg. of sales @ Tk. 1.00.

Fixed marketing and Administrative expenses Tk. 3,00,000.

Standard costing system is used.

Required :

(i) Income statement under absorption costing.

(ii) Income statement under variable costing system.

Solution : ৮৭ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–12**

The following particulars are available from the books of M Ltd. for the year ended 2020 :

Sales 1,50,000 units @ Tk. 30 per unit, Beginning finished goods 24,000 units, Ending finished goods 35,000 units. Manufacturing cost :

Variable cost per unit Tk. 12, Fixed factory overhead Tk. 1,60,000, Normal capacity 1,60,000 units, variable marketing and administrative expenses Tk. 1.50 per unit of sales. Fixed administrative expenses Tk. 3,00,000.

Required :

(a) Income statement under direct costing.

(b) Income statement under absorption costing.

(c) Reconciliation statement.

Solution : ৬৩৯ পৃষ্ঠার **Question No-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2012 No-11; 2014 No-11; (Finance) 2018 No-12**

Ahmed Ltd/Sobuj Ltd. uses absorption costing system. The president has recently heard about variable costing. However, he supplies you the following data :

Normal capacity	1,000 units
Beginning inventory	200 units
Production	900 units

Sales 1,000 units @ Tk. 10.00 per unit
Materials costs Tk 2.00 per unit
Labour Tk 2.00 per unit
Variable overhead Tk 1.00 per unit
Fixed manufacturing expenses Tk 2,700
Fixed selling expense Tk 1,300
Unfavourable variable cost variance from standard Tk. 200
Required : Income statement under :
(a) Absorption costing method;
(b) Variable costing method.

Solution : ৯০ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 12**

The following particulars are available from the books of a manufacturing concern :

Normal capacity	20,000 units
Standard variable manufacturing cost per unit	Tk. 15
Variable marketing expenses per unit sold	Tk. 5
Fixed factory overhead	Tk. 40,000
Fixed marketing expenses	Tk. 25,000
Selling price per unit	Tk. 30

Selected data for the company's operations in the last year were as follows :

Beginning inventory	2,000 units
Sales	17,500 units
Production	18,000 units

Required : Prepare income statement using variable costing and absorption costing.

Solution : ৯২ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2022 No–14**

Tasnia traders sells its product for Tk. 30 per unit. The standard cost per unit is :

Account titles	Taka
Direct materials	10
Direct labour	7
Variable factory overheads	3
Fixed factory overheads (normal capacity of 20,000 units)	4
Total	24

At the beginning, there were 5,000 units of products on hand. During the year 15,000 units were produce and 15,000 units were sold. Per year selling expenses Tk. 20,000 plus 5% on sales.

Required :
(i) Prepare an income statement using variable costing
(ii) Prepare an income statement using absorption costing.

Solution : ৯৩ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

বিবিএ (হিসাববিজ্ঞান) সাজেশন্স (অনার্স ৩য় বর্ষ)

oo **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2015 No–12(b)**

ABC Ltd. produces a product whose related information for the year ended December, 31-2014 is as follows :

Account Titles	Data
Beginning inventory	20,000 units
Normal capacity	1,00,000 units
Production during the year	90,000 units
Ending inventory	30,000 units
Direct material cost per unit	Tk. 20
Direct labour cost per unit	Tk. 5
Variable factory overhead per unit	Tk. 6
Selling price per unit	Tk. 40
Annual fixed factory overhead	Tk. 1,00,000
Fixed administrative and marketing expense	Tk. 60,000
Variable administrative and marketing expense	Tk. 40,000

Required : Prepare an income statement for the year using direct costing and absorption costing method.

Solution : ৯৪ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–13**

Tamanna Manufacturing Company uses a standard cost system in accounting for its product which it sells for Tk. per unit. The standard cost per unit is :

Direct materials	Tk. 7
Direct labour	Tk. 6
Variable factory overhead	Tk. 3
Fixed factory overhead (Based on normal capacity of 70,000 units)	Tk. 4
Total cost	Tk. 20

All variances are closed to cost of goods sold.

On June 1, 2020 there were 5,000 units of product on hand. During June 60,000 units were produced and 50,000 units were sold. Cost incurred during June were :

Direct materials	Tk. 4,10,000
Direct labour	Tk. 3,90,000
Variable factory overhead	Tk. 2,00,000
Fixed factory overhead	Tk. 2,50,000
Variable marketing overhead	Tk. 75,000
Fixed marketing overhead	Tk. 80,000

Required :

(a) Prepare income statement under variable costing.

(b) Prepare income statement under absorption costing.

Solution : ১০৩ পৃষ্ঠার **Problem-28** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–13**

The below data pertain to August 2023 :

Beginning inventory	**Nill**
Units sold	5,000 units
Units produced	8,000 units
Sales per unit	Tk. 24
Direct manufacturing cost (per unit)	Tk. 10
Fixed factory overhead per unit	Tk. 3.50
Fixed factory overhead (total)	Tk. 28,000
Selling and distribution expense (all are fixed)	Tk. 10,000

Required :
(a) Prepare two income statements using direct costing and absorption costing
(b) Prepare reconciliation statement between two methods.

Solution : ১০৬ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-15** ***Year Question & Solution*** **⊙ N.U. BBA (Hon's) 3rd (Mgt.) 2017 No–14(b); 2021 No–5; D.U. BBA (Hon's) 3rd (Mgt.) 2017 No–11; MBA Preli (Acc.) 2014 No–2(b)**

Jamuna Corporation produced 24,000 units (normal capacity) of product during the year: 20,000 units were sold @ Tk. 22 per unit. Cost of production was as follows :

Direct materials	Tk. 60,000
Direct labour	60,000
Factory overhead :	
Variable cost	1,20,000
Fixed cost	96,000

Marketing and administrative expenses for the year Tk. 70,000, all are fixed.
(i) Income statement under absorption costing and variable costing.
(ii) Reconciliation statement.

Solution : ১০৭ পৃষ্ঠার **Problem-31** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-16** ***Year Question & Solution*** **⊙ N.U. BBA (Hon's) 3rd (Acc.) 2017 No–13; MBA Preli (Acc.) 2019 No–13**

Sabuj Limited is comparing its present absorption costing practice with variable costing methods.
An examination of its records revealed the following information :
Budgeted production 45,000 units
Budgeted fixed factory overhead Tk. 1,35,000
Fixed marketing and admin expense Tk. 15,000
Sales price per unit Tk. 20
Standard variable manufacturing cost per unit Tk. 8
Variable marketing expense per unit Tk. 2
For the year the following data are available :
Actual production 40,000 units
Actual sales 35,000 units
Finished stock inventory (opening) 2,000 units
Unfavorable variance from standard cost Tk. 8,000.
Required :
(1) Income statements
(2) Reconciliation statements.

Solution : ১১১ পৃষ্ঠার **Problem-33** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-17** ***Year Question & Solution*** **⊙ N.U. BBA (Hon's) 3rd (Old) 2013 No–03; (Management) 2017 No–07**

The following data are for the East-West Ltd.

Account Titiles	January	February
Sales (in units)	4,000	6,000
Production (in units)	8,000	2,000
Sales price per units	Tk. 20	Tk. 20
Fixed factory over head (Total)	Tk. 24,000	Tk. 24,000
Direct manufacturing cost per unit	Tk. 10	Tk. 10
Fixed factory overhead per unit	Tk. 3	Tk. 3
Selling and administrative expense	Tk. 8,000	Tk. 8.000

You are required to prepare comparative income statement based on absorption costing method.

Solution : ১১৭ পৃষ্ঠার **Problem-37** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2019 No–12**

ACI Ltd. has the following data for the first two years storing in 2017. Basic production data at standard cost are as follows :

Account Titles	Taka
Direct materials	13.00
Direct labor	15.00
Variable overhead	2.00
Fixed overhead (Tk. 15,00,000 ÷ 1,50,000 units)	10.00
Total standard factory cost per unit	40.00

Selling price per unit Tk. 50.

Fixed selling and administrative expenses Tk. 6,50,000. Sales commission 5% of sales value. Output, sales and inventory (in units):

Accounts titles	2017	2018
Beginning inventory		30,000
Production	1,70,000	1,40,000
Sales	1,40,000	1,60,000
Ending inventory	30,000	10,000

Required :

(a) An income statement for two years under variable costing.

(b) An income statement for two years under absorption costing.

(c) A reconciliation statement for two years.

Solution : ১১৯ পৃষ্ঠার **Problem-39** নং প্রশ্নের উত্তর দেখুন।

Chapter 04 Activity Based Costing

(কার্যভিত্তিক উৎপাদন ব্যয় হিসাব)

Part-A : Brief Questions & Solutions :

01. What is ABC costing? [২০২৪ সালে যা এসেছিল]
Or, What is activity based costing (ABC)?
[N.U. 2013 (Old); 2020, 2024 (Acc.); 2019 (Mgt.); 2016(Finance)]
Ans. *Activity based-costing refers a costing system by identifying individual activities as the fundamental cost objects.*

oo **02. What is activity?**
Ans. The condition in which things are happening or being done is called activity.

03 **What is activity cost pool?** 99%
[N.U. 2018, 2022 (Acc.)]
Ans. Set of costs which are incurred when certain operations are performed within the organization is called activity cost pool.
[রকেট স্পেশাল সাজেশন: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo **04. What is activity measure?**
Ans. An activity measure is an allocation base of indirect costs in an activity based costing system.

ooo **05. What is cost object?** 99%
Ans. A cost object is often a product or department for which costs are accumulated or measured.

oo **06. What is meant by cost allocation?**
Ans. Cost allocation is the process of identifying, aggregating and assigning costs to cost objects.

ooo **07. What is cost allocation base?**
Ans. Cost allocation base is cost driver or activity. base on which base indirect costs allocate to the products or services.

ooo **08. What is traditional costing system?**
Ans. Traditional costing is a simple costing that allocates overhead costs to the products or services by applying broad average costing using a single indirect cost pool.

oo **09. What is activity cost driver?**
Ans. A cost driver is the unit of an activity that causes the change in activity's cost.

ooo **10. What is Activity-based management (ABM)?** 99%
[N.U. 2016 (Acc)]
Ans. Activity based management (ABM) is a method of identifying and evaluating activities that a business performs using activity based costing to carry out a value chain analysis of a re-engineering initative to improve strategic and operational decisions in an organization.

ooo **11. What is activity based budget?** *[N.U. 2016 (Mgt.)]*
Ans : Budgets that focus on the budgeted cost of activities required to produce and sell products and service are called activity based budget.

Part- B & C (Short & Broad Questions):

✪✪✪ 01. **What do you mean by ABC costing? 99%**
[Ans : হিসাববিজ্ঞান ইজিপ্রেস (৩য় খণ্ড) বইয়ের 123 Page 4.01 No.]
Or, Definition of activity based Costing.

✪✪ 02. Discuss the objective of ABC costing. **[Ans : 125 Page 4.05 No.]**

✪✪✪ 03. **Definition of ABM (Activity based management). 99% [Ans : 126 Page 4.09 No.]**
Or, What do you mean by ABM.

✪✪ 04. "Integrity is more important for business professionals than it is for business students."—Do you agree? Explain. *[Test Exam : Satkhira city, Tangail Karmichael College; N.U.(Acc)- 2016]*

✪✪✪ (05) **What are the characteristics of activity based costing? 99% [Ans : 124 Page 4.02 No.]** *[N.U.(Acc)- 2022]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

4. *Show the differences between simple costing system and activity based costing system.*
[Ans : Page No. 125; Ques. No. 4:07] *[Test Exam- Azizul, Rajshahi, Gopagonj, Magara College; N.U. 2024 (Acc)]*
Or, Difference between traditional costing and activity based costing.
Or, Show the difference between TC costing and ABC costing.

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 01 & 05 No. Question in are most important for the Examinee of 2025 ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

✪✪✪ **Problem-01** ***Year Question & Solution*** **⊙ N.U. BBA (Hon's) 3rd (Acc.) 2018 No–6**
SB Ltd. produced 10,000 digital cameras in the most recent quarter and has determined that total activity costs were Tk. 50,00,000 of materials costs, Tk. 12,00,000 of labor costs, Tk. 25,00,000 of inspection costs inspection costs and Tk. 5,00,000 of packing costs. It takes 30 minutes of labor to produce each camera, inspections are done for 25% of all cameras produced and cameras are packaged individually. What are the rates for each of there activities? (Assume the company has decided to switch from a volume-based system to an activity based system.)

Solution : হিসাববিজ্ঞান ইজিপ্রেস (৩য় খণ্ড) বইয়ের ১২৭ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-02** ***Year Question & Solution*** **⊙ N.U. (MBA Final) Acc. 2016 No– 15**
Tamarin Company produces speakers models A and model B. Both products pass through two producing departments. Model A's production is much more labor-intensive than model B. Model B is also the more popular of the two speakers. The following data have been gathered for the two products :

Particular	Model A	Model B
Units produced per year	30,000	3,00,000
Prime costs	Tk. 2,00,000	Tk. 20,00,000
Direct labor hours	1,40,000	3,00,000
Machine hours	20,000	2,00,000
Production runs	40	60
Inspection hours	800	1,200
Maintenance hours	10,000	90,000

Overhead costs :	Taka
Setup costs	3,60,000
Inspection costs	2,80,000
Machining	3,20,000
Maintenance	3,60,000
Total	13,20,000

Required :

(a) Compute the overhead cost per unit for each product using a plant wide rate based on direct labor hours.

(b) Compute the overhead cost per unit for each product using activity-based costing.

Solution : ১২৮ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. (MBA Final) Acc. 2017 No– 12**

ABC Company Produce two types of Xerox machine. Data related to the two products are given below :

Annual production in units	25,000	50,000
Direct material costs	Tk. 6,00,000	Tk. 12,00,000
Direct manufacturing labour cost	Tk. 2,00,000	Tk. 4,00,000
Direct manufacturing labour hour	5,000	10,000
Machine hours	25,000	50,000
Number of production run	100	100
Inspection hours	1,000	500

Total manufacturing overhead costs are :

	Taka
Machine cost	15,00,000
Set up cost	4,80,000
Inspection cost	4,20,000

Required :

(a) Compute the manufacturing overhead cost per unit for each product.

(b) Compute the manufacturing cost per unit for each product.

Solution : ১৩০ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-04** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 3rd (Acc.) 2017 No–15; N.U. BBA (Hon's) 3rd (Acc.) 2022 No–15**

Joy company makes two products :

Product	Direct labour hours per unit	Production
P	0.80	10,000 units
Q	0.40	40,000 units

Additional information for the company :

(a) P requires Tk. 32 in direct materials per unit and Q requires Tk. 18.

(b) The direct rate is Tk. 15 per hour.

(c) P is more complex to manufacture than Q and they require special processing.

(d) The ABC system has the following activity cost pools:

Activity cost pool and activity measure	Estimated overhead cost (Tk.)	P	Q	Total
Machine set up (No. of set up)	72,000	100	300	400
Special processing (Machine houres)	2,00,000	5,000	–	5,000
General factory (Direct labour hours)	8,16,000	8,000	16,000	24,000

Required :

(i) Compute the activity rate for each activity cost pool.

(ii) Determine the unit cost of each product according to the ABC system including direct materials and direct labour.

Solution : ১৩২ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০ Problem-05 *Year Question & Solution* ⊙ N.U. (MBA Final) Acc. 2014 No– 13

Dell Company manufactures two models of Ultra-high fidelity speakers, the X-200 model and X-99 model. Data regarding the two products follow:

Product Name	Direct Annual Total Direct	Product	Direct Annual Total Direct
	Labour Hours (DLHs)	Production	Labour Hours
X-200	1.8 DLH per unit	10,000 units	18,000 DLHs
X-99	0.9 DLH per unit	60,000 units	54,000 DLHs
			72,000 DLHs

Additional information about the company follows :

(a) Model X-200 requires Tk. 72 indirect materials per unit and model X-99 requires Tk. 50.

(b) The direct labour rate is Tk. 10 per hour.

(c) The company has always used direct labor-hours as the base for applying manufacturing overhead cost to products.

(d) Model X-200 is more complex to manufacture than model X-99 and requires the use of special equipment.

(e) Because of the special work required in (d) above, the company is considering the use of activity based costing to apply manufacturing overhead cost to products. Three activity cost pools have been identified as follows :

Activity Cost Pools	Activity Measure	Total Cost
Machine setups	Number of setups	Tk. 3,60,000
Special processing	Machine hours	1,80,000
General factory	Direct labor-hours	12,60,000
		Tk. 18,00,000

	Expected Activity		
Activity measure	Model X-200	Model X-99	Total
Number of setups	50	100	150
Machine-hours	12,000	0	12,000
Direct labor-hours	18,000	54,000	72,000

Required :

(1) Assume that the company continues to use direct labor-hours as the base for applying overhead cost to products.

(a) Compute the predetermined overhead rate.

(b) Compute the unit product cost of each model.

(2) Assume that the company decides to use activity based costing to apply overhead cost to products.

(a) Compute the activity rate for each activity cost pool and determine the amount of overhead cost that would be applied to each model using activity based costing system.

(b) Compute the unit product cost of each model.

(3) Explain why overhead cost shifted from the high-volume model to the low volume model under activity-based costing.

Solution : ১৩৩ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

Chapter 05 Cost Volume Profit Relationships

(ব্যয় পরিমাণ মুনাফার সম্পর্ক)

Part-A : Brief Questions & Solutions :

ooo 01. **What is cost?** *[N.U. 2016 (Finance)*
Or, What do you mean by cost?
Ans : Cost is a measurement of monetary terms of the amount of resources used for a specific purpose such as production of goods and services.

oo 02. **What is cost driver?**
Ans : Any output measure of resource and activities is called cost driver.

ooo 03. **What is production cost?**
Ans : Production cost is the cost of sequence of operation which begins with supplying materials, labour and service and ends with primary packing of the product.

ooo 04. **Define marginal cost?**
[N.U. 2013, 2015 (Old); 2018 (Finance)]
Ans : The additional cost resulting from producing and selling one additional unit is called marginal cost.

05 **What is safety margin?** 99%
[Test Exam : Kabi Nazrull, Titumir & Dendra College; N.U. 2012, 2014(Old); 2016, 2019, 2022 (Acc.); 2016, 2018 (Mgt.); 2019, 2021 (Finance)]
Or, What do you mean by margin of safety?
Or, Define margin of safety?
Ans : The excess of budgeted (or actual) sales over the break-even volume is margin of safety.
[রকেট স্পেশাল সাজেশন: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 06. **What is markup?** *[N.U. 2017 (Acc.); 2019 (Mgt.)]*
Ans : Markup is the difference between selling price and cost. It's usually expressed as a percentage of sales price.

ooo 07. **What is break even point?** 99%
[Test Exam : Titumir & Suharawardy College; N.U. 2015 (Old); 2020 (Mgt.)]
***Or,* Define BEP (break even point).**
Ans : The break-even point can be defined as the point where total revenue equals total cost and as the point where total contribution margin equals total fixed costs.

ooo 08. **Write down the formula of BEP.** 99%
[N.U. 2021 (Acc.)]
Or. Write down the formula of break-even point.
Ans : $BEP = \frac{\text{Fixed cost}}{\text{Contribution per unit}}$

oo 09. **Write a formula of BEP in taka.**
[N.U. 2021 (Mgt.)]
Ans. $BEP\ (Tk.) = \frac{\text{Total fixed cost}}{\text{P/V Ratio}}$

ooo 10. **What do you mean by break-even chart?**
Or, What is break-even chart? 99%
[N.U. 2018 (Acc.); 2020 (Finance)]
Ans : The relationship between revenue, cost and level of activity in an organization presented in graphic form and determine BEP from it, is called breack-even chart.

ooo 11. **Write down the formula of BEP ratio.**
[N.U. 2013 (Old)]
Ans : (i) BEP Ratio = 100% – M/S Ratio,
(ii) $BEP\ Ratio = \frac{BEP}{Sales}$

ooo 12. **What is C/M ratio?**
[N.U. 2012 (Old); 2016, 2021 (Finance)]
Or, What is the contribution margin ratio?
Ans : The contribution margin ratio is the ratio of the total contribution margin to total sales revenue. This term is synonymous with profit-volume ratio.

ooo 13. **Write down any one application of C/M ratio.**
[N.U. 2014 (Old); 2019 (Mgt.); 2021 (Acc.)]
Or, Write the formula of CM ratio.
Or, Write down the formula of contribution margin ratio. 99%
Ans : Contribution margin ratio
$= \frac{\text{Contribution}}{\text{Sales}} \times 100$

oo 14. **What is the equation method?**
Ans : A method of computing the break-even point that relies on the equation is called equation method.

oo 15. **Write down the formula of margin of safety?**
Ans : (i) Margin of safety = Sales – BEP Sales;
(ii) $\text{Margin of Safety} = \frac{\text{Profit}}{\text{P/V Ratio}}$

ooo 16. **What is profit volume ratio?**
Ans : The ratio that is used to analysis the relation between contribution and sales is known as profit volume ratio.

17 **What is sales mix?** 99%
[N.U. 2016, 2018, 2020 (Mgt.)]
Ans : The relative combination in which a company's product are sold. Sales mix is computed by expressing the sales of each product as a percentage of total sales.
[রকেট স্পেশাল সাজেশন: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 18. **What is operating leverage?**

Ans : The potential use of fixed operating costs to magnify the effects of change in sales on the firm's earnings before interest and taxes is called operating leverage.

ooo 19. **What is cost drive?**

Ans : The cost drive of a variable cost is the level of activity or volume whose change causes proportionate changes in the variable cost.

ooo 20. **What is the elaboration of CVP?** *[N.U. 2017 (Acc.)]*

Ans. CVP = Cost Volume Profit.

ooo 21. **How contribution margin is derived?** 99% *N.U. 2017 (Acc.)]*

Or, Write down the formula of contribution.

Ans. Contribution margin = Total actual or expected sales - Total variable cost.

ooo 22. **What is contribution margin (CM)?** *[N.U. 2019 (Acc.); 2021 (Mgt.)]*

Ans. The amount remaining from sales revenues after all variable costs have been deducted is called contribution margin.

ooo (23) **Write down the formula of profit volume ratio.** 99% *[N.U. 2020 (Acc.)]*

Ans. P/V ratio = $\frac{\text{Contribution per unit}}{\text{Selling price per unit}}$

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 24. **What is marginal revenue?** *[N.U. 2016 (Finance)]*

Ans. The additional revenue resulting from the sale of an additional unit is called marginal revenue.

ooo 25. **What is CVP analysis?** 99% *[N.U. 2021 (Finance)]*

Ans. CVP analysis defined as an analysis of the effect of value changes upon costs and upon profits

Part- B & C (Short & Broad Questions) :

oo 01. What is CVP analysis? [**Ans :** রেটাভেক্সা- হিসাববিজ্ঞান হাইলাইটস (৩য় খণ্ড) বইয়ের **136 Page 5.01 No.**] *[N.U. 2005]*
Or, Definition of CVP analysis.

oo 02. State the underlying assumptions of CPV analysis. [**Ans : 141 Page 5.22 No.**] *[N.U BBA (Hon's) (Acc)-2015]*

ooo 03. **What are the assumptions of cost volume profit analysis?** 99% [**Ans : 140 Page 5.18 No.**] *[N.U 2012, 2014 (Accounting & Management)-2018; (Finance)-2017; D.U. (Acc)-2017]*
Or, State the assumptions of cost-volume-profit analysis.
Or, Discuss the assumptions of cost-volume-profit analysis.

ooo (04) **How does cost volume profit analysis help in management decision-making?** 99%
[**Ans : 140 Page 5.17 No:**] *[N.U (Acc)-2015, 2022; (Finance)-2018]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo 05. **What are the principal assumption underlying the break even analysis?** 9% [**Ans : 139 Page; 5.12 No.**] *[N.U (Acc) 2022]*

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 03 & 04 No. Questions in are most important for the Examinee of 2025. .....Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

ooo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2022 No–06**
Tasin Ltd. has a profit of Tk. 35,000, fixed cost Tk. 40,160 and break even point Tk. 1,25,500. Find out the amount of sales.

Solution : রেটাভেক্সা হিসাববিজ্ঞান হাইলাইটস (৩য় খণ্ড) বইয়ের ১৪৫ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. - 2018 No–08**
Find out the amount of sales from the following data :

Fixed cost	Tk. 25,000
Profit	Tk. 18,000
Break-even sales	Tk. 75,000

Solution : ১৪৫ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2016 No–7**

Mumu Company has fixed expenses of Tk. 90,000 with sales at Tk. 3,00,000 and a profit of Tk. 60,000. If in the next period, the company suffered a loss of Tk. 30,000, calculate the sales volume.

Solution : ১৪৬ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–5**

The selling price of a product is Tk. 20 per unit and contribution margin is 25%. If Tk. 15,000 profit is made by selling 15,000 units of a product, calculate break even sales Tk.

Solution : ১৪৬ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০০ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 04**

Sales Tk. 12,00,000, CM ratio 40%, and MS ratio 25%.

Required :

(a) Variable cost in Taka,

(b) Profit in Taka.

Solution : ১৪৭ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–8**

Rakib company for the last year disclosed an M/S ratio of 20% and C/M ratio of 60%. Fixed cost Tk. 60,000 Calculate : (i) BEP (Tk.) (ii) Profit (iii) CM.

Solution : ৬৩৭ পৃষ্ঠার **Question No-08** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc.- 2017 No–04**

The operating results of Najia Enterprise are given below :-

Sales for 2015 and 2016 are Tk. 5,00,000 and Tk. 8,00,000 respectively. Profit for 2015 is Tk. 1,00,000 and for 2016 is Tk. 1,75,000.

Required : (i) P/V ratio; (ii) Fixed cost.

Solution : ১৪৯ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2022 No–05**

TAZ Ltd. costs of producing 2,500 units in January and 3,400 units in February were Tk. 21,830 and Tk. 24,980 respectively. The Co. plans to produce 2,000 units in March. How much would be the cost of production in March?

Solution : ১৪৯ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–06**

Operation of 'Z' Company for the last year disclosed an M/S ratio 20% and a C/M ratio 50%. Fixed cost Tk. 40,000.

Required :

(i) Break even sales;

(ii) The amount of profit

(iii) The contribution margin.

Solution : ১৫০ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–15**

The following data is extracted from the books of a company :

Annual sales	Tk. 2,00,000
Margin of safety ratio	30%
Profit volume ratio	40%

Required : Ascertain–

(a) Break-even in sales in taka
(b) Margin of safety in taka
(c) Annual profit
(d) Sales required to earn a profit of Tk 40,000
(e) Profit or loss on sale of Tk. 1,50,000.

Solution : ১৫৪ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2022 No–5**

You have been supplied with the following information :

Net sales	Tk. 6,00,000
Variable cost	Tk. 3,00,000
Fixed costs	Tk. 1,80,000

Required :

(i) Break even sales in taka
(ii) Margin of safety ratio
(iii) Contribution margin ratio
(iv) Present net profit in taka

Solution : ১৫৭ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

ooo (১১%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2015 No–5; (Management) 2021 No–9**

The following figures for profit and sales are obtained from the account of 'X' Co. Ltd./ XYZ Co :

Year	Sales	Profit
	Tk.	Tk.
2013	20,000	1,000
2014	30,000	3,000

Calculate :

(i) Profit volume ratio;
(ii) BEP;
(iii) Profit when sales Tk. 40,000;
(iv) Sales to earn a profit of Tk. 5,000.

Solution : ১৫৯ পৃষ্ঠার **Problem-28** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

XX **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–13**

Sakib Ltd. furnishes you the following data relating to the year 2024 :

	First half of the year	Second half of the year
Sales	Tk. 50,000	Tk. 80,000
Total cost	Tk. 40,000	Tk. 55,000

Assuming that there is no change in prices and variable cost and fixed cost are incurred equally in the two half year periods, calculate for the year 2024 :

(i) P/V ratio; (ii) Fixed cost; (iii) BEP sales; (iv) Sales required earning profit Tk. 30,000; (v) Profit when annual sales Tk. 1,00,000.

Solution : ৬৪১ পৃষ্ঠার **Question No-13** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old)-2014 No–15; Finance 2021 No–13**

XYZ Company Ltd. /A company collected the following data for the second quarter of the year :-

Months	Sales (Tk.)	Costs (Tk.)
April	70,000	56,000
May	77,000	59,990
June	85,000	64,550

Required :

(i) Variable costs per unit in Taka;
(ii) The fixed costs;
(iii) The break-even point in Tk.;
(iv) July profit if sales Tk. 80,000.

Solution : ১৬১ পৃষ্ঠার **Problem-31** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Management- 2017 No–08**

Swift Corporation has a single product whose selling price is Tk. 120 and whose variable cost is Tk. 80 per unit. The company's monthly fixed expense is Tk. 50,000.

Required :

(i) Using the equation method, solve for the unit sales that are required to earn a target profit of Tk. 10,000.
(ii) Using the contribution margin approach, solve for the taka sales that are required to earn a target profit of Tk. 15,000.

Solution : ১৬২ পৃষ্ঠার **Problem-32** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2022 No–16**

The following data are obtained from the record of Jamuna Company :

Accounts titles	Taka	Taka
Sales (8,000 units @ Tk 25 each)		2,00,000
Direct materials consumed	80,000	
Variable overhead	25,000	
Labour	35,000	
Fixed overhead	36,000	1,76,000
Net profit		24,000

Required :

(i) The number of units by selling which the Co. will neither loss nor gain anything.
(ii) The sales needed to earn profit of 20% on sales.
(iii) The extra unit which should be sold to obtain the present profit it is proposed to reduce the selling price by 20%.
(iv) The selling price to be fixed to bring down its break-even point to 1,000 units under present condition.

Solution : ১৬৪ পৃষ্ঠার **Problem-35** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০০ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2016 No–09; 2018 No–09**

Scientific Ltd. produces and sales calculator. Variable cost and selling price per calculator are Tk. 600 and Tk. 800 respectively. The fixed cost per year is Tk. 36,000. You are required to calculate :

(i) Break even sales in unit and taka.
(ii) Sales to earn a profit of Tk. 60,000 (before tax).
(iii) Sales to earn a profit of Tk. 45,000 after tax (the tax rate is 40%).

Solution : ১৬৫ পৃষ্ঠার **Problem-36** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2016 No–13(b)**

The Kohinoor Chemical Company has sales of Tk. 1,20,000 and margin of safety 40% and P/V ratio 30% during the current year. The company expects that next year the fixed cost and the sales price will increase. As a result the margin of safety and the P/V ratio will change to 35% and 40% respectively.

Required :

(i) The present break even sales;
(ii) The amount of current profit;
(iii) The break even sales next year;
(iv) The amount of profit next year.

Solution : ১৬৭ পৃষ্ঠার **Problem-38** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Mgt. 2017 No–12; (Finance) 2022 No–13; (MBA Preli.) (Acc.) 2018 No–13**

A company plans to produce 5,00,000 units at a variable cost of Tk. 20 each. The fixed costs are expected to be Tk. 20,00,000. The selling price is fixed to give 25% profit on cost.

Required :

(i) P/V ratio;
(ii) Break-even point;
(iii) P/V ratio and break-even point if the selling price is increased by 20%.
(iv) The number of units required to be sold at the reduced selling price to obtain an increase of 20% over the budgeted profit.

Solution : ১৬৮ পৃষ্ঠার **Problem-40** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2018 No–13**

The following data is extracted from the books of Padma Company Ltd.

Account Titles	Taka	Taka
Sales 1,00,000 units		8,00,000
Purchase price of products	3,00,000	
Commission (10% on sales)	80,000	3,80,000
Contribution margin		4,20,000
Fixed cost		3,00,000
Profit before tax		1,20,000

The managing director of the company has some questions for you to answer. Each situation is independent of other :

(i) What is the company's break-even point?
(ii) What unit sales would give you a pre-tax profit of Tk. 1,83,000?
(iii) If the company could sales 1,00,000 units and wants a pre-tax profit of Tk. 2,10,000. What price per unit would it have to charge?
(iv) If the purchase price of the product increased by 10%, what volume would be required to earn an after tax profit of Tk. 77,400 (Tax rate 40%)?

Solution : ১৬৯ পৃষ্ঠার **Problem-41** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Management) 2019 No–14(b); 3rd (Acc.) 2020 No– 13; (MBA Preli.) (Management) 2015 No–5(b)**

Sakil Co. Ltd./Otobi Ltd./Glorious Ltd. manufactures and sells a single product. The company's sales and expenses for last quarter is as follows :

Accounts titles	Total Taka	Per unit Taka
Sales	9,00,000	60
Less : Variable cost	3,60,000	24
Contribution margin	5,40,000	36
Less : Fixed cost	4,32,000	
Net operating income	1,08,000	

Required :

(i) What is company's CM ratio?
(ii) What is quarterly break-even-point in units sold and in sales taka?
(iii) Company's margin of safety in taka and in percentage term.
(iv) How many units would be sold each quarter to earn a pre-tax profit of Tk. 1,80,000?
(v) If company can sell only 15,000 units but wanted to earn a pre-tax profit of Tk. 1,80,000, then what would be the selling price per unit?
(vi) If sales increase by Tk. 1,00,000 per quarter and there is no change in fixed expense, by how much would you expect net operating income to increase?

Solution : ১৭৫ পৃষ্ঠার **Problem-46** নং প্রশ্নের উত্তর দেখুন।

২০২৪ সালের পরীক্ষায় যা এসেছিল

⊗⊗ **Problem-22** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–17**

A company has three product lines, all are produced in one factory. Management is deep concerned with product C which has consistently shown a net loss. The income statement is given below :

Particulars	A	B	C	Total
Sales	5,00,000	4,00,000	1,00,000	10,00,000
Less : Variable cost	2,95,000	2,80,000	75,000	6,50,000
Contribution margin	2,05,000	1,20,000	25,000	3,50,000
Fixed expenses (Salaries, depreciation etc.)	1,65,000	90,000	45,000	3,00,000
Net income (Loss)	40,000	30,000	(20,000)	50,000

Required :

(i) Whether product C will be dropped when only alternative is to drop product C or to continue with it. Assume that fixed expenses of product includes Tk. 20,000 supervision salary. Assume further that the total assets invested will not be affected by the decision.
(ii) Drop product C, keeping the supervision and using the vacant facilities to product A to satisfy its expanding demand by an additional sales of Tk.1,00,000. Which alternative is more profitable.

Solution : ৬৪৬ পৃষ্ঠার **Question No-17** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-23** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2019 No–12**

Tania Ltd. using 50% of capacity produces 30,000 units last year for market. Of the total cost Tk. 90,000 was variable cost and Tk. 70,000 was fixed cost. Due to trade depression, the company has to reduce the price from Tk. 8 to 6 per unit during the current year. The company has also decide to increase production to 50,000 units and to sell the excess production to foreign market at Tk. 5.00 per unit. Selling expenses for the foreign sales will be Tk. 0.50 per unit. Fixed cost will increase by Tk. 5,000.

Required :

(i) Profit or loss at home and foreign market during the current year.
(ii) Break-even sales during the current year.

Solution : ১৭৭ পৃষ্ঠার **Problem-48** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-24** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–12**

In 2022 margin of safety, contribution margin ratio and margin of safety ratio of Korotowa Ltd. were Tk. 40,000, 25% and 20% respectively. In 2023 Meghna Ltd. wants to sales 24,000 units by increasing it sales by 20%. It is expected that in 2023, unit selling price will increase by 10%, variable cost per unit will increase by Tk. 0.20 and fixed cost will decrease by 10% Tax rate 40%.

Required :

(a) Calculate break even sales, actual sales and profit % for 2022 and 2023.
(b) Determine margin of safety ratio for 2023.
(c) Determine the target sales to earn after tax profit Tk. 14,400.

Solution : ১৭৯ পৃষ্ঠার **Problem-50** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-25** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2015 No–13(b)**

Cost details of a company are given below :

Account Titles	Taka
Sales	12,00,000
Direct mateirals	2,50,000
Direct labour	2,00,000
Fixed factory overhead	2,50,000
Variable factory overhead	1,50,000
Fixed selling expense	70,000
Variable selling expense	80,000
Administrative expense	40,000

Required : Compute the BEP sales and show it graphically.

Solution : ১৮১ পৃষ্ঠার **Problem-51** নং প্রশ্নের উত্তর দেখুন।

Chapter 06 Relevant Costs for Decision Making

(সিদ্ধান্ত গ্রহণে প্রাসঙ্গিক ব্যয় প্রক্রিয়া)

Part-A : Brief Questions & Solutions :

01. What is relevant cost? [২০২৪ সালে যা এসেছিল]
[N.U. 2012, 2014(Old); 2016, 2019, 2024 (Acc.); 2016, 2020 (Mgt.); 2016, 2021 (Finance)]
Ans : Relevant cost are those expected future cost that differ among alternative course of action.

ooo **02. What is imputed cost?** 99% *[N.U.2023Acc.)]*
Ans : A cost that is incurred by virtue of using an asset instead of investing it or undertaking an alternative is called imputed cost.

ooo **03. What is avoidable cost?**
Ans : A cost that can be avoided by not producing particular goods is called avoidable cost.

oo **04. What is unavoidable cost?** *[N.U. 2015 (Old); 2017 (Mgt.)]*
Ans : A cost that cannot be influenced at the business unit level but it is controllable at the corporate level is called unavoidable cost.

oo **05. What is irrelevant cost?**
Ans : An irrelevant cost is a cost that will not changed as the result of a management decision.

ooo **06. What is controllable cost?**
Ans : Controllable cost is the cost which can be influences by the action of a specified member of the undertaking.

07. What is cross-subsidization? 99% *[N. U. 2023 (Acc.)]*
Ans : Cross-subsidization occurs when the profits from one product, service or customer segment are used to cover the losses lower prices of another.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo **08. What is historical cost?**
Ans : When an asset is given up or a liability is incurred to acquire an assets, it is called historical cost.

ooo **09. What do you mean by decision making?** 99% *[N.U. 2022 (Acc.) 2021 (Mgt.)]*
Or, What is decision making?
Ans : Decision making is choosing the best among the alternatives.

ooo **10. Define quantitative factors decision making.**
Ans : Quantitative factors are those that can be expressed in numerical terms.

ooo **11. What are qualitative factor?**
Ans : Qualitative factors are those that cannot be expressed in numerical terms but are considered for decision making.

oo **12. What is uncontrollable cost?**
Ans : Uncontrollable cost is the costs which cannot be influenced by the action of a specified number of the undertaking.

ooo **13. What is cost indifference point?** 99% *[N. U. 2023 (Acc.)]*
Ans : The cost indifference point is the level of activity at which the total costs of two alternative options are equal.

ooo 14. **What is abandoning cost?**

Ans : Abandoning costs are the costs of retiring altogether a fixed asset from use.

ooo 15. **Define relevant range.** 99%

[N.U. 2022(Acc.); 2017 (Mgt.)]

Or, What is relevant range?

Ans. The relevant range is the range of activity within which the assumption about variable and fixed cost behavior are effective.

oo 16. **Elaborate : ACCA.** *[N.U. 2018 (Finance)]*

Ans : Association of Chartered Certified Accountants.

(17) **What is relevant cash flow?** 99%

[N.U. 2022, 2023 (Acc.); 2020 (Finance)]

Ans. The relevant cash flows is future incremental cash flow arising from the decision being made. This means that the cash flows are only relevant if they are future, incremental costs.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo 18. **What is special order?** *[N. U. 2021 (Acc.)]*

Ans. A one time order that is not considered part of the company's normal ongoing business is called special order.

Part- B & C (Short & Broad Questions):

oo 01. What is sunk cost? [Ans : খোন্দকার হিসাববিজ্ঞান ইন্টারপ্রেস (৩য় খণ্ড) বইয়ের 187 Page 6.03 No.]

Or, Definition of sunk cost?

oo 02. (a) What is opportunity cost? **[Ans : 187 Page 6.02 No.]**

Or, What do you mean by opportunity cost?

(b) "All relevent costs are futurs costs but all future costs are not relevent cost." Do you agree and why? **[Ans : 191 Page 6.16 No.]** *[NU. 2017, 2019 (Mgt.), 2020 (Finance)]*

ooo 03. **What are the procedures of different decision making by using incremental cost analysis?** 99%

[Ans : 188 Page 6.05 No.] *[N. U. (Acc)-2018]*

oo 04. How an investment proposal is evaluated according to different discounted cash-flow techniques?

[Ans : 190 Page 6.14 No.] *[N.U. (Acc)-2015]*

(05) **Why relevant cost is an important managerial tool? Explain with example.** 99%

[Ans : 191 Page 6.19 No.] *[NU. 2013, D.U. (Mgt.)-2016, 2023 (Acc)]*

Or, Why is relevance costing an important managerial decision tool? Explain with examples?

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 03 & 05 No. Question in are most important for the Examinee of 2025 ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

ooo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 08**

Delta Company is considering expansion of its plant capacity. Fixed costs amount to Tk. 4,20,000 and are expected to increase by Tk. 1,25,000 when plant expansion is completed. The present plant capacity is 80,000 units per year. Capacity will increase by 50% with the expansion. Variable costs are currently Tk. 6 per unit and are expected to go down by Tk. 1 per unit with the expansion. The current selling price is Tk. 20 per unit and expected to remain same under either alternative. Which alternative is better expand or not to expand? Why?

Solution : খোন্দকার হিসাববিজ্ঞান ইন্টারপ্রেস (৩য় খণ্ড) বইয়ের ১৯৩ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2021 No–14 (b)**

Rahman Ltd. has the capacity of producing 20,000 units per month. At the moment, they are producing and selling 12,000 units per month at Tk. 20 per unit. Cost per unit are as follow :

Direct materials	Tk. 8
Direct labour	4
Variable factory overhead	2
Fixed factory overhead	1
Fixed administrative expense	1
Variable selling expense	1

Recommend whether or not the company should accept a special order for 8,000 units at Tk. 17 per unit.

Solution : ১৯৫ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–8**

The Sagor Company is using 15,000 units of part L-5 as a component to assemble one of its products. It costs the company to produce it internally having itemized expenditures as follows :

Direct material	Tk. 1,50,000
Direct labor	Tk. 1,55,000
Variable overhead	Tk. 90,000
Fixed overhead	Tk. 3,00,000

An outside vendor has just offered to supply the part for Tk. 35 per unit. If the company stops producing this part, $\frac{1}{4}$ of the fixed overhead would be avoided.

Should the company make or buy this part?

Solution : ১৯৪ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 16**

Sovon Ltd. manufactures 40,000 parts of specialized products. At this level of activity the cost per parts is as follows :

Accounts titles	Taka
Direct materials	5
Direct labour	10
Variable overhead	2
Fixed overhead	8
Total cost per parts	25

An outside supplier has offered to sell 40,000 parts each year to Sovon Ltd. for Tk. 22 per parts. If Sovon Ltd. accepts this offer, the facilities now being used to manufacture the parts could be rented to another company at an annual rental of Tk. 80,000. However, the company has determined that Tk. 6 per unit of the fixed overhead being applied to the parts would continue even the parts are purchased from outside.

Required : Should the parts be purchased from outside or continue to produce in the factory?

Solution : ১৯৭ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old)-2015 No–14(b); (Finance) 2018 No–16(b); 2019 No–06; (Mgt) 2021 No–7; (MBA Prell.) (Mgt) 2019 No–08**

Turjo Company Ltd./ABC Ltd. is a manufacturing concern produces Parts-B in its own factory. The relevant cost of the product as raw material Tk. 20, direct labor Tk. 80, variable overhead Tk. 50 and fixed overhead Tk. 50. Company needs' 2,000 units of Parts-B.

A supplier offers to sell Parts-B to ABC Ltd. at a cost of Tk. 160 per unit. If the offer is accepted fixed overhead can be saved Tk. 35 per unit. Should the offer be accepted?

Solution : ১৯৭ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2017 No–09; (Finance) 2022 No–6**

Faisal Chemical Ltd. turns out as single product. The highest capacity of company is 6,00,000 kgs. Presently the firm produces 4,00,000 kgs. It has received a foreign offer for 80,000 kgs @ Tk. 13 per kg. The company sells Tk. 17 per kg. The standard cost sheet per kg are given below :

Account Titles	Taka
Direct materials	6.00
Direct labour	3.50
Variable overhead	3 25
Fixed overhead	1.75
	14.50

Required : Should the company accept the offer?

Solution : ১৯৮ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2018 No–14(b)**

Moon Company manufactures 40,000 parts of specialised products. At this level of activitiy the cost per part is as follows :

Account Titles	Taka
Direct materials	8
Direct labor	7
Variable overhead	2
Fixed overhead	8
Total cost per parts	25

An outside supplier has offered to sell 40,000 parts each year to the company for Tk. 22 per parts. If the company accepts this offer, the facilities now being used to manufacture the parts could be rented to another company at an annual rental of Tk. 80,000. However, the company has determined that Tk. 6 per unit of the fixed overhead being applied to the parts would continue even the parts are purchased from outside.

Required : Should the parts be purchased from outside or continue to the produce in the factory?

Solution : ১৯৮ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2022 No–07**

Noor Co. has been producing and selling 30,000 units of its product per month with the following total costs:

Accounts titles	Taka
Direct materials	90,000
Direct labour	75,000
Variable factory overhead	60,000
Fixed factory overhead	80,000
Variable marketing and administrative expenses	45,000
Fixed marketing and administrative expenses	50,000

The normal sales price is Tk. 17 and plant capacity is 40,000 units. The company has received an offer from a special customer who would like to buy exactly 7,000 units of product for Tk. 9 per unit. Marketing and administrative expenses related to this special order would be Tk. 4,000.

Required : Should Noor Co. accept this special order?

Solution : ১৯৫ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

oooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. (MBA Preli.) (Acc.) 2017 No- 7(c); 3rd (Finance) 2022 No–17**

Mizan Ltd. manufactures 4,000 units of a product during a year. Cost of the product is as follows :

Accounts titles	Taka
Direct materials	86,000
Direct labour	50,000
Overheads	45,000

60% of productive overheads are fixed. Administrative and selling expenses are Tk. 26,000 and Tk. 20,000 respectively. Administrative expenses are entirely fixed but 30% of selling expenses are variable. A foreign customer has given a proposal to the company to buy 1,000 units. The company sells the product at Tk. 63 but the foreign buyer offers Tk. 50 as the price of the product, No additional cost would be incurred for the increase sales except Tk. 2,000 as delivery charges.

Should the offer of the foreign customer be accepted?

Solution : ২০২ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2017 No–14**

Modumoti and Company produces 24,000 units in two shifts, the costs of which are furnished below :—

Account Titles	Taka
Direct material	3,60,000
Direct labour	2,40,000
Factory overhead	1,44,000
Administrative expense	96,000
Selling expense	84,000

75% of factory overhead, 100% of administrative expense and 70% of selling expense are fixed.

Now the management of Modumoti and company is considering to start 3rd shift in order to increase production proportionately. If the third shift is started, cost of all raw materials will be reduced by 2%, direct labour rate of the third shift will increase by 20%, fixed factory overhead will increase by Tk. 12,000 and the selling price is to be reduced from Tk. 150 to Tk. 145.

You are required to give suggestion to management authority whether the 3rd shift should be started.

Solution : ২০৪ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

Chapter 07 Budgeting and Budgetary Control
(বাজেটিং এবং বাজেটীয় নিয়ন্ত্রণ)

Part-A : Brief Questions & Solutions :

০০ **01. What is budgeting?**

Ans : Budgeting is the co-ordination of financial and non-financial planning of satisfy the organization goal.

০০০ **02. What is budget?**

Ans : Budget is a plan showing a company's objectives and how management intends to acquire and use resource to attain those objectives.

০০০ **03. What is cash budget?** 99%

[N.U. 2012 (Old); 2016, 2020 (Acc.); 2016, 2018, 2020 (Mgt.); 2016, 2018 (Finance)]

Ans : Cash budget is plan indicating expected inflows (receipts) and outflows (disbursements) of cash.

০০০ **04. What is master budget?** 99%

[Test Exam : B M Suhrawardy, Debendra College; N.U. 2013 (Old); 2016, 2022 (Acc.) ;2017, 2019, 2021 (Mgt.); 2021 (Finance)]

Ans : The master budget is a set of interrelated budget, that constitute a plan of action for a specified time period.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

০০ **05. What is financial budget?** ***[N.U. 2018 (Acc.)]***

Ans : Financial budget is the part of master budget that focuses on the effects of the operating budget and other plans will have on cash balances.

oo 06. **What is fixed budget?**

Ans : Fixed budget is a budget which is designed to remain unchanged irrespective of the level of activity actually attained.

07 **What is rolling budget?** 99% *[N. U. 2023 (Acc.)]*

Ans : Rolling budget is a form of master budgets that simply adds a month in future as the month just ended is dropped.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo 08. **What is sales budget?** *[N.U. 2017 (Acc.)]*

Ans : A sales budget is detailed schedule showing the expected sales of the budget period typically it is expressed in both price and units of product.

ooo 09. **What do you mean by production budget?**

Ans : Production budget is a detailed plan showing the number of units that must be produced during a period in order to meet sales and inventory needs.

ooo 10. **What is budget cycle?** *[N.U. 2021 (Mgt.)]*

Ans. All the activities of budgetary control system are to be done in circulating nature. This circulating nature of budgeting is called budget cycle.

11 **Mention the two non-discounting techniques of capital budgeting.** 99% *N. U. 2023 (Acc.)]*

Ans : Two non-discounting techniques of capital budgeting are (i) Payback period method & (ii) Average rate of return method.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 12. **What is capital budgeting?** *[N.U. 2016 (Mgt.)]*

Ans.: Capital budgeting refers to the planning of expenditure whose returns are expected to spread over a series of years.

ooo 13. **What is payback period?** *[N.U. 2020 (Finance)]*

Ans : The payback period is the length of time that it takes for a project to recover its initial cost from the net cash inflows that it generates.

ooo 14. **What is cost of capital?** 99%

[N.U. 2022 (Acc.); 2021 (Finance)]

Ans : The appropriate interest rate is the cost of the finance provided to the company by shareholders and other investors, as investment projects have to earn a return at least sufficient to pay the interest on the funds they utilize.

oo 15. **What is NPV?** *[N.U. 2017 (Acc.); 2016 (Mgt.)]*

Ans : Capital investments usually earn return that extends over fairly long period of time.

oo 16. **Define profitability index?**

Ans : A measurement of the profitability of an investment computed by dividing the net present value by the initial cost of investment.

ooo 17. **Write the formula of net profitability index?**

Ans : Net Profitability Index = PI – 1.

ooo 18. **What is IRR?** 99% *[Test Exam : B.M, M.M. Suhrawardy & Debendra College; N.U. 2015 (Old); 2018, 2021 (Mgt.); 2021 (Acc.)]*

Or, What is meant by IRR?

Ans : IRR is the discount rate at which the present value of expected cash inflows from a project equals the present value of expected cash outflows of the project.

ooo 19. **What is meant by budgetary control?** 99%

Or, What is budgetary control? *[N.U. 2016, 2018 (Acc.); 2017, 2020 (Mgt.); 2016 (Finance)]*

Ans. The exercise of control in the organization with the help of budgets is known as budgetary control.

20. What is zero based budget? [২০২৪ সালে যা এসেছিল] [N.U. 2016, 2018, 2024 (Acc.); 2017, 2019 (Finance)]

Or, What is ZBB?

***Ans.** The budget which comes from zero is called zero based budget.*

oo 21. **What is capital rationing?** *[N.U. 2016, 2020, 2021 (Acc.); 2017, 2019 (Mgt.); 2017, 2019 (Finance)]*

Ans. Capital rationing refers to the choice of investment proposals under financial constraints in terms of given size of capital.

22. What is participatory budgeting? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Acc.)]

***Ans :** Participatory budgeting is a budgeting approach in which managers and employees at different levels actively participate in the preparation of the budget.*

oo 23. **What is budget centre?** *[N.U. 2018 (Acc.)]*

Ans. Budget centre is the segment of the enterprise in respect of which budget is first prepared.

ooo 24. **What is operating budget?** *[N.U. 2019 (Acc.)]*

Ans. The operating budget focuses on the income statement and its supporting schedules for in an organization with no sales revenues on budgeted expenses and supporting schedules.

25 **Define working capital?** 99%

[N.U. 2015 (Old); 2018, 2020, 2022 (Mgt.); 2021 (Acc.)]

Or, What is working capital?

Ans. Working capital is the amount of money a business has available to funds its day to day operation.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 26. **Elaborate IFAC.** 99%

[N.U. 2017 (Finance); 2023)]

Ans : International Federation of Accountants.

oo 27. **What is sales forecasting?** *[N.U. 2017 (Finance)]*

Ans. Sales forecasting is a statement of what the amount or value of a company's sales is likely to be in the future, based on information available now about the market; past sales etc.

ooo 28. **Elaborate : TQM, CAT.** *[N.U. 2021 (Mgt)]*

Ans. TQM = Total Quality Management. and **CAT** = Certified Accounting Technician.

oo 29. **What is proforma statement?**

[N.U. 2017 (Finance)]

Ans. The master budget/proforma statement is a set of interrelated budget, that constitute a plan of action for a specified time period.

ooo 30. **What is strategic budget?** 99%

[N.U. 2020 (Finance)]

Ans. The budget adjusted for strategic planning is called strategic budget.

31. A project's NPV at 10% discount rate is zero (0), what will be the IRR? [২০২৪ সালে যা এসেছিল] *[N.U. 2024 (Acc.)]*

***Ans :** A project's NPV at 10% discount rate is zero (0), then the IRR is 10%.*

Part-B & C (Short & Broad Questions):

oo 01. (a) What is the meaning of budget and budgetary control?

[Ans : সাজেশন্স হিসাববিজ্ঞান ইজিপ্লাস (৩য় খণ্ড) বইয়ের 208+210 Page 7.01+7.09 No.]

Or, Define of budget and budgetary control? *[Test Exam : Dhaka College; N.U.BBA 2005, 2009, 2010]*

(b) What are the differences between budget and budgetary control? **[Ans : 211 Page 7.13 No.]**

[Test Exam : Dhaka, Eden & Debendra College; N.U. 2005, 2007, 2008, 2010, 2014; (Acc)-2015, 2017; (Mgt.)-2016; D.U.-(Acc)-2017]

Or, Distinguish between budget and budgetary control.

Or, State differences between budget and budgetary control.

oo (02) **Discuss the components of a master budget.** 99% **[Ans : 213 Page 7.21 No.]**

[Test Exam : Dhaka, Magara, Daokandi, Nzaimuddin, Pabna College;N.U. (Acc)-2015, 2018]

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo 03. Distinguish between budget and forecast. **[Ans : 211 Page 7.14 No.]**

[N.U. BBA (Hon's)-2013; (Acc)-2016; (Mgt.)-2017]

ooo 04. **Distinguish between budget and standard.** 99% **[Ans : 217 Page 7.33 No.]** *[N.U (Acc)-2018]*

ooo 05. Discuss the essentials elements of a good budgetary control. **[Ans : 215 Page 7.25 No.]**

[Test Exam : Pangsha, B.L, B.M, Satkhira, Sadat College; N.U. 2013 (old)-(Acc)-2016]

oo 06. Define sales budget, Describe the factors considered in preparing sales budget.

[Ans : 216 Page 7.28+7.29 No.] *[N.U. (Acc)-2018]*

oo 07. What are the differences between strategic planning and long-range planning? **[Ans : 225 Page 7.71 No.]**

[N.U. (Finance)-2017]

ooo 08. **(a) Describe the importance of cash budget.** 99% **[Ans : 217 Page 7.37 No.]**. *[N.U. (Acc)-2016, 2021]*

(b) Distinguish between cash budget and cash flow statement. 99% **[Ans : 218 Page 7.39 No.]**

[N.U. (Acc)-2021]

oo 09. Distinguish between zero based budget and traditional budget: **[Ans : 225 Page 7.72 No.]**

[N.U. (Acc)-2021]

ooo (10) **Describe briefly steps in budgetary control.** 99% **[Ans : Page No 191; Ques. No 6.20]**. *[N.U. (Acc)-2022]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

2. Distinguish between a master budget and a sales forecast. **[Ans : Page No. 634 Question No. 2]** *[N.U. (Acc)-2024]*

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 02, 08 & 10 No. Questions in are most important for the Examinee of 2025..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

1. Purchase Budget

oo **Problem-01** ***Basic Question & Solution***

The sales director of a manufacturing company reports that next year he expects to sell 54,000 units of a certain product. The production manager consults the store-keeper and casts his figures as follows :

Two kinds of raw materials, A and B are required for manufacturing the product. Each unit of the product requires 2 units of A and 3 units of B. The estimated opening balances at the commencement of the next year are :

Finished product – 10,000 units, A – 12,000 units, B – 15,000 units.

The desirable closing balances at the end of the next year are:

Finished product – 14,000 units, A – 13,000 units, B– 16,000 units.

Draw up a quantitative chart showing the materials purchases budget for the next year.

Solution : বিশ্লেষণ হিসাববিজ্ঞান ইজিপ্রেস (৩য় খণ্ড) বইয়ের ২২৭ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

2. Sales Budget

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Old-2015 No–15(b); 3rd (Finance) 2019 No–15**

PQR Ltd./ABC Ltd. produces and sales two types of products named 'K' and 'M'. In 2017, the sales of the products were as follows :

Regions	Product A	Product B
East	20,000 units	30,000 units
West	25,000 units	35,000 units

The standard selling price of product 'K' and 'M' are Tk. 5 and Tk. 8 respectively. The following points are to be considered in preparation of sales budget in 2018.

(a) The selling price of 'K' should be reduced by 20% to increase the volume of sales by 30%.

(b) Due to scarcity of raw materials of product 'M', 20% of the plant in which 'M' is produced will remain idle for which selling price of 'M' should be increased by 25%.

Prepare a sales budget for 2018.

Solution : ২২৮ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

3. Labour Budget

oo **Problem-03** ***Basic Question & Solution***

For Everly Company, units to be produced are 5,000 in 1st quarter and 6,000 in 2nd quarter. It takes 1.5 hours to make a finished unit and the expected hourly wage rate is Tk. 14 per hour. Prepare a direct labour budget by quarters for the 6 months ending June 30, 2013.

Solution : ২২৮ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

4. Overhead Budget

oo **Problem-04** ***Basic Question & Solution***

Hardin Company is preparing its manufacturing overhead budget for 2013. Relevant data consist of the following.

Units to be produced (by quarters) : 10,000, 12,000, 14,000, 16,000.

Direct labour : Time is 1.5 hours per unit.

Variable overhead costs per direct labour hour : Indirect materials Tk. 0.70; indirect labour Tk. 1.20 and maintenance Tk. 0.50.

Fixed overhead costs per quarter : Supervisory salaries Tk. 35,000, depreciation expense Tk. 16,000 and maintenance Tk. 12,000

Instructions : Prepare the manufacturing overhead budget for the year showing quarterly data.

Solution : ২২৯ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

5. Materials Budget

oo **Problem-05** ***Basic Question & Solution***

Ortiz Company has 1,600 pounds of raw materials in its December 31, 2012, ending inventory. Required production for January and February of 2013 are 4,000 and 5,500 units, respectively. Two pounds of raw materials are needed for each unit, and the estimated cost per pound is Tk. 6. Management desires an ending inventory equal to 20% of next month's materials requirements. Prepare the direct materials budget for January.

Solution : ২২৯ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

6. Production Budget

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 09**

Gold Star Ltd. has budgeted sales of its products for the next four months as follows :

Month	Sales in units
April	50,000
May	75,000
June	90,000
July	80,000

The company is now in the process of preparing a production budget for the second quarter. Past experience has shown that end-of-month inventory levels must equal 10% of the following month's sales. The inventory at the end of March was 5,000 units.

Required : Prepare a production budget for the second quarter.

Solution : ২৩০ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

২০২০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc.- 2018 No–07**

BD Ltd. will start business from 1st July 2017. Followings are the budgeted sales for 2017 :

Account titles	July	Aug.	Sept.	Oct.	Nov.
Estimated sales (in units)	15,000	18,000	12,000	13,500	13,500

The company has a policy of carrying an end of month inventory at 100% of the following month's sales plus 50% of the second following month's sales. Prepare a production budget for the first three months.

Solution : ২৩১ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2016 No–15(c)**

The production manager of XYZ company desires to maintain an inventory of raw material equal to the budgeted production needs of the next 2 months. Each unit of product takes 10 kgs of raw materials. Inventory of finished goods is usually maintained at one and half (i.e.1.5) times of the following month's sales. The forecast sales of 6 months of next year, in units are as follows :

January	3,600	April	4,800
February	4,200	May	5,600
March	3,200	June	3,800

On December 31 of last year, there were 80,000 kgs of raw material on hand and 5,600 units of finished goods.

You are required to prepare :

(i) Budget for production.

(ii) Budget for purchase of raw material in units for as many months as possible.

Solution : ২৩৩ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

7. Cash Budget

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–08**

ABC Company's sales budget for the third quarter of 2020 is given below :

Particular	July (Tk.)	August (Tk.)	September (Tk.)	Total (Tk.)
Budgeted sales	3,00,000	4,00,000	3,50,000	10,50,000

25 percent of a month's sales are collected in the month of sales, 65 percent is collected in the month following sales and remaining 10 percent is collected in the second month following sale. May sales totalled Tk. 2,50,000 and June sales totalled Tk. 2,50,000.

Required : Show the schedule of cash collections from sales by month and in total third quarter.

Solution : ২৩৪ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

ooo (১১%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Old 2014 No–06; Finance 2021 No–09**

Sales proceeds of a Co. are collected as follows:

40% in the month of sales

40% in the second month and

20% in the third month.

Sales for the month of June Tk. 2,40,000. Accounts receivable are Tk. 1,20,000 in May, 31, 2015. Consisting of Tk. 30,000 from April sales and Tk. 90,000 from May sales.

Show the schedule of cash collection from sales for the month of June' 2015.

Solution : ২৩৫ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Old-2012 No–05; Finance-2017 No–08; 2022 No–08**

ABC Company's /Farzana company's sales budget for the third quarter of 2016 is given below :-

	July	August	September
Budgeted sales	3,50,000	4,50,000	2,50,000

20% of a monthly sales are collected in the month of sales, 70% is collected in the month following sales and remaining is collected in the second month following sales. May sales totalled Tk. 2,15,000 and June sales totalled Tk. 2,70,000.

Required :

(i) Show the schedule of cash collection from sales by month.

(ii) Show the account receivable as on closing data.

Solution : ২৩৬ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's)Acc.-2018 No–15**

Considering the following income statement, prepare a cash budget of the Silver Co. for the month of June 2017.

The Silver Company Ltd.

Budgeted income statement for the month of June, 2017

Accounts titles	Taka	Taka
Sales		2,40,000
Inventory, May 31	40,000	
Purchase	1.60,000	
	2,00,000	
Inventory, June 30	30,000	1,70,000
Gross margin		70,000
Operating expenses :		
Wages	25,000	
Utilities	5,000	
Advertising	7,000	
Depreciation	3,000	
Office expense	4,000	
Insurance	3,000	47,000
Operating income		23,000

The cash balance on May 31, 2017 is Tk. 40,000. Sales proceeds are collected as follows :
40% in the month of sales; 40% in the 2nd month and 20% in the 3rd month. Accounts receivables are Tk. 1,20,000 in May 31, 2017 consisting of Tk. 30,000 from April's sales and Tk. 90,000 from May's sales. Accounts payable on May 31, 2017 are Tk. 1,20,000. The company pays 25% of the purchase during the month of purchase and the remainder during the following month. All operating expenses requiring cash are paid during the month of recognition. However, insurance is paid annually in December. Confine you analysis go given data. Ignore income tax.

Solution : ২৩৯ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2014 No–13; (Acc.) 2016 No–16(b); (Finance) 2020 No– 05; 4th (Acc.) 2016, 2019 No–12; D.U. 3rd (Acc.) 2016 No–15**

Lamiya/Modern/Puspita/Rose manufacturing furniture company has the following forecast data available:

Account Titles	September Tk.	October Tk.
Cash sales	40,000	60,000
Sales on credit	77,500	95,000
Cash purchase	20,000	20,000
Purchase on credit	1,00,000	80,000
Purchase discounts	6,000	3,000
Accounts payable beginning	10,000	12,000
Accounts payable ending	12,000	9,000
Cash operating expenses	46,500	10,000

Net sales on credit are collected 50% in the month of sales, 40% in the following and 10% in the second following months. Such sales in July and August were Tk. 1,00,000 and Tk. 1,20,000 respectively. The estimated September 1 cash balance is Tk. 13,000.

Prepare a cash budget for September and October.

Solution : ২৪০ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020. No– 14**

A company making stock in the first quarter of the year is assisted by its bankers with cash facilities. The following are the relevant budget figures :

Month	Credit sales	Credit purchases	Wages and expenses
	(Taka)	(Taka)	(Taka)
November	1,20,000	83,000	10,000
December	1,28,000	96,000	10,000
January	1,80,000	1,40,000	11,000
February	1,16,000	1,20,000	10,000
March	84,000	40,000	12,000

Given the following further information you are required to prepare a cash budget for the quarter January to March. Other information are as follows :

(i) Budgeted cash at bank on January 1, Tk. 25,000.

(ii) Credit terms of sales are payable by the end of the month following the month of supply. On an average one-half of sales are paid on the due date, while the other half are paid during the next month. Creditors are paid during the month following the month of supply.

(iii) Wages and expenses are paid twice in the month of 1st and 16th respectively.

Solution : ২৪৫ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–15**

From the information given below prepare a cash budget of X Ltd. for 1st quarter 2024. Cash balance 1st January Tk. 30,000. Monthly wage and Salary Tk. 10,000, debenture interest payable in March 2024 Tk. 10,000. Other information are given below :

Details	December	January	February	March
Cash sales	80,000	1,40,000	1,50,000	1,21,000
Credit sales	1,00,000	80,000	1,40,000	1,20,000
Purchase	1,60,000	1,80,000	2,40,000	1,80,000
Other payment	16,000	20,000	22,000	20,000

Terms of purchases and sales :

(i) 10% of the purchases are paid promptly and the remaining in 30 days basis.

(ii) Credit sales are collected 50% in the month of sales are made and 50% in the following month, collection from credit sales are subject to 5% discount if payment is received during the month of sales and 2.5% discount if payment is received in the month following.

Solution : ৬৪৩ পৃষ্ঠার **Question No-15** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.-2022 No–17**

From the following information prepare a cash budget for the three months commencing 01 June, 2022. The bank balance on that date is Tk. 1,10,000 :

Details	April (Tk.)	May (Tk.)	June (Tk.)	July (Tk.)	August (Tk.)
Sales	88,500	84,000	93,000	72,000	82,000
Selling overhead	3,250	4,100	3,710	3,210	3,600
Purchases	37,000	40,000	39,000	39,000	35,000
Wages	8,000	8,400	8,800	6,000	9,600
Factory overhead	5,680	5,920	2,440	5,880	6,000
Administrative overhead	2,500	2,750	2,480	2,600	2,520

Others information :

(i) A sales commission of 2% on sales is due 2 months after sales in payable in addition to the above selling overhead.
(ii) Plant purchases June 01 for Tk. 38,000, payable on delivery.
(iii) Building purchases in the month of June for Tk. 80,000, payable in two half yearly installment, the 1st in the month of July.
(iv) A dividend of Tk. 20,000 will be paid in July
(v) Sales and purchases are made 10% on cash. Credit allowed by suppliers is one month. Debtors are allowed two months credits.
(vi) Time lag-in-payments wages and all other overheads is 15 days.

Solution : ২৫০ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

8. Capital Budget

oo **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc.-2017 No–08**

A Machine is purchased for Tk. 1,00,000 having an economic life 5 years. The cash inflows from this machine for five years are Tk. 10,000, Tk. 15,000, Tk. 50,000, Tk. 40,000 and Tk. 60,000. If the tax is 30%, What is cashflow after taxes (CFAT)?

Solution : ২৫২ পৃষ্ঠার **Problem-31** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance-2018 No–15**

R.K Ltd. is considering on investment proposal to install new equipment facility. The project will cost Tk. 65,000 having a life expectancy of 5 years. The project has no salvage value. The firm uses straight line method of depreciation. The company's tax rate is 40%. The essential cash flow before depreciation and tax (CFBT) from the proposed investment proposal are as follows :

Year	CFBT
	Taka
1	10,500
2	18,000
3	12,500
4	19,000
5	14,000

You are required to calculate the net present value (NPV) at 10% and give your opinion about project.

Solution : ২৫৪ পৃষ্ঠার **Problem-34** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2020 No– 07; (Management) 2021 No–8**

A project initial cost is Tk. 80,000. Its expected life is 5 years and discount rate is 10%. Cash inflows are given below :

Year	Amount
	(Taka)
1	24,000
2	30,000
3	26,000
4	20,000
5	28,000

You are required to calculate discounted payback period.

Solution : ২৫৬ পৃষ্ঠার **Problem-37** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc,) 2020 No– 07**

Meghna Ltd. is considering to purchase a new machine. The machine will cost Tk. 3,00,000. The cash inflows per year would be Tk. 80,000. The machines economic life is 10 years and cost of capital is 12%. The tax rate of the company is 45%. Calculate NPV and suggest whether the machine should be purchased or not. Assume PVIFA for Tk. 1.00 @ 12%, 10 years is 5.6502.

Solution : ২৫৭ পৃষ্ঠার **Problem-39** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–07**

'Y' Manufacturing Ltd. is examining the profitability of invating in new project. The following ormation is available in respect of the project:

Investment	Tk. 12,70,000
Residual value at the end of life	Tk. 70,000
Life of the project	20 years
Annual profit of the project (before tax)	Tk. 1,80,000

Tax rate 50%

Required :

(i) Payback period (ii) Net present value (iii) Average rate of return.

Solution : ২৬০ পৃষ্ঠার **Problem-43** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

XX **Problem-22** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–9**

XY Ltd. is examining the profitability of investing in new project. The following information is available in respect of the project :

Investment	Tk. 20,00,000
Salvage value	Tk 1,00,000
Life of project	20 years
Annual profit of the project (before tax) is	Tk. 3,50,000

Calculate : (i) PBP (ii) ARR.

Solution : ৬৩৭ পৃষ্ঠার **Question No-09** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-23** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–16**

A company is considering a project with required an investment Tk. 4,00,000 and also need Tk. 50,000 as a working capital. After life time the estimated salvage value Tk. 30,000. Tax rate is 50%. The company uses straight line depreciation and proposed project has cash flow before tax (CFBT) Tk. 1,70,000 per annum

Determine the following :

(a) Pay back period
(b) Average rate of return
(c) Net present value (Assume cost of capital is 15%)
(d) Internal rate of return

Solution : ২৬১ পৃষ্ঠার **Problem-44** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-24** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Management-2018 No–14**

A company is considering an investment proposal which requires an initial cash outlay of Tk. 50,000. The investment will have a life of 5 years. The project will have no salvage value at the end of the life. The company uses straight line method at depreciation. The company's tax rate is 40%. The estimated cash flow before tax are follows :-

Year	Cash flows (Taka)
1	12,000
2	14,000
3	16,000
4	18,000
5	20,000

The PV factors are : 0.9091; 0.8264; 0.7513; 0.6830 and 0.6209

Required :

(i) Pay back period (PBP)
(ii) Net present value (NPV)
(iii) Profitability index (PI)
(iv) Net profitability index (NPI)
(v) Super life

Solution : ২৬৭ পৃষ্ঠার **Problem-50** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-25** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2019 No–16(b); Finance-2022 No–12**

Crown Ltd. is considering a proposal to install a new machine. The project will cost Tk. 1,00,000 and its expected life is five years. The estimated cash-inflows before tax (CFBT) of the project are as follows :

Year	1	2	3	4	5
CFBT	25,000	30,000	35,000	50,000	40,000

The tax rate is 50% and charges depreciation on straight line basis :
Required : (i) PBP; (ii) ARR; (iii) NPV(10% discount rate).

Solution : ২৬৮ পৃষ্ঠার **Problem-51** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-26** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance-2017 No–13**

A company purchased a machine two years ago. In the meant time, most modern machines are available in the market. The compahy wants to examine whether the replacement of the old machine by a new one would be profitable. The particulars relating to the two machines are given below :—

	Old Machine (Tk.)	New Machine (Tk.)
Cost	1,40,000	1,80,000
Book value	1,00,000	—
Realizable value	60,000	—
Realizable value after 5 years	10,000	40,000
Overhauling cost after 3 years	25,000	—
Annual operating cost (excluding depreciation)	70,000	25,000

Cost of capital is 18% and tax rate is 40%. Should the machine be replaced?

Solution : ২৭২ পৃষ্ঠার **Problem-54** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-27** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–15**

A manufacturing company is considering an older machine which was fully depreciated for tax purposes with a new machine costing Tk. 50,000. The new machine will be depreciated over its 10 years life. It is estimated that the new machine will reduce labour costs by Tk. 10,000 per year. The Management believes that there will no change in expenses and revenues of the firm due to the machine. The company requires an after tax return on investment of 10% and tax rate 35%. The company's income statement for the current year is given for other information's:

Particulars	Tk.	Tk.
Sales		5,00,000
Less : costs :		
Material	1,50,000	
Labour	2,00,000	
Factory and administrative	42,000	
Depreciation	40,000	(4,32,000)
Net income before tax		68,000
Less : Taxes (35%)		(23,800)
Earning after taxes		44,200

Should the company buy the new machine? Assume that the company follows straight line method and same tax rate is allowed for purchasing new machine.

Solution : ২৭১ পৃষ্ঠার **Problem-53** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-28** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Management) 2020 No– 17**

The details of investment proposal of ASM Company are given below :

Initial cash outlay (Tk.)	1,00,000
Expected life	5 years
Salvage value (Tk.)	10,000
Working capital (Tk.)	20,000

Cash inflows before tax :

Year	Taka
1	25,000
2	30,000
3	35,000
4	40,000
5	50,000

The company's cost of capital is 10% and corporate tax rate is 40%. The project will be depreciated on the basis of straight line. Evaluate the project on the basis of

(i) Payback period (PBP)
(ii) Net present value (NPV)
(iii) Profitability index (PI)
(iv) Internal rate of return (IRR).

Solution : ২৭৮ পৃষ্ঠার **Problem-60** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-29** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's)Acc.- 2016 No–14(b)**

Raj Chemical Company is considering a project which requires an investment of Tk. 40,000. The estimated salvage value is zero. Tax rate 50%, cost of capital 10%. The company uses straight line depreciation and the proposed project cash-flows before tax (CFBT) as follows :—

At the end of year	1	2	3	4	5
CFBT (Tk.)	10,000	10,000	15,000	15,000	25,000

Determine payback period and IRR if cost of capital 10%.

Solution : ২৮০ পৃষ্ঠার **Problem-62** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-30** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Management-2017 No–17**

You are a financial analyst of a company. The director of capital budgeting has asked you to analyze two proposed capital investments, project A and B. Each project has a cost of Tk. 10,000 and the cost of capital for each project is 12%. The project's expected net cash-flows are as follows :

Expected net cash-flows

Year	Project A	Project B
0	(Tk. 10,000)	(Tk. 10,000)
1	6,500	3,500
2	3,000	3,500
3	3,500	3,000
4	1,500	2,500

(i) Which project should be accepted if they are independent?
(ii) Which project should be accepted if they are mutually exclusive?

Solution : ২৮১ পৃষ্ঠার **Problem-63** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-31** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Management) 2019 No–16**

From the following data calculate :

(i) Pay back period.

(ii) NPV.

(iii) IRR for the following projects.

Assume a required rate of return 14% and 40% tax rate. The company has a policy of charging depreciation on straight line method.

	Project-X	Project-Y
Initial outlay	Tk. 1,00,000	Tk. 1,40,000
Salvage value	Nil	Tk. 20,000
Expected life	5 years	5 years

Earnings before depreciation and taxes

Year	Taka	Taka
1	25,000	45,000
2	30,000	35,000
3	20,000	30,000
4	20,000	30,000
5	15,000	25,000

Which of the projects would you select and why?

Solution : ২৮৪ পৃষ্ঠার **Problem-65** নং প্রশ্নের উত্তর দেখুন।

Chapter 08 Flexible Budget

(নমনীয় বাজেট)

Part-A : Brief Questions & Solutions :

01. What do you meant by flexible budget? [২০২৪ সালে যা এসেছিল][N.U. 2013, 2015 (Old); 2017, 2020, 2024 (Acc.); 2018 (Mgt.) 2018-(Finance)]

Or, What is flexible budget?

Ans : A budget which is designed to changed in accordance with the level of activity actually attained is called flexible budget.

oo 02 **What is static budget?** 99% *[N.U. 2021 (Finance)]*

Or, What do you mean by static/fixed budget?

Ans : A budget which is designed to remain unchanged irrespective of the level of activity actual attained is called static/fixed budget.

রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

ooo 03. **What is budget variance?** 99%

[N.U. 2018 (Acc.); 2019 (Finance)]

Ans : A budget variance is an accounting term that describes instances where actual costs are either higher or lower than the standard or projected costs.

ooo 04. **Write down steps of preparing flexible budget.**

Ans : (a) Fix the range of activity level, (b) Cost analysis, (c) Select the activity level, (d) Prepare the budget at each activity level.

oo 05. **What is cost behaviour analysis?**

Ans : Cost behaviour analysis is the study of how specific costs respond to changes is the level of business activity.

ooo 06. **Mention feature of variable cost.**

Ans : There are two variable cost:

(i) Variable cost in total amount as the activity level changes. (ii) Unit variable cost is fixed.

ooo 07. **Give two examples of semi-variable costs.** 99%

[N.U. 2016 (Mgt.); 2017, 2020 (Finance)]

Ans : Telephone expenses (Fixed or land phone). Light and Electricity expenses.

ooo 08. **What is semi-variable cost?** 99%

[N.U. 2017 (Finance.); ; 2020 (Mgt.)]

Ans. Semi-variable cost is the mixture of variable cost and fixed cost.

ooo 09. **Write two example of variable costs.**

Ans : There are two variable cost:

(i) Direct labour and (ii) Direct material.

Part- B & C (Short & Broad Questions):

○○ 01. Discuss the Features/Characteristies of flexible budget? *[D.U. (Mgt.)-2016]*

[Ans : হিসাববিজ্ঞান ইজিপ্রেপ (৩য় খণ্ড) বইয়ের 287 Page 8.04 No

○○○ 02. Discuss the disadvantages/Demerits/Limitation of flexible budget? **[Ans : 287 Page 8.03 No.]**

○○○ 03. **What is the difference between flexible budget and fixed budget? 99% [Ans : 288 Page 8.08 No.]**

[Test Exam : Dhaka College; N.U. 2012, Accounting-2015, 2017; (Finance)-2018]

Or, Show the differences between fixed budget and flexible budget.

○○○ (04) **What are the advantages of flexible budget? 99% [Ans : 287 Page 8.02 No.]**

[Test Exam : Siddashsori College; N.U. Accounting-2016, 2017, 2021; (Mgt.)-2018]

Or, Discuss the advantage of flexible budget.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

○○○ (05) **Discuss the some common usage of flexible budget. 99% [Ans : Page No 289; Ques. No 8.14]** *[N.U. 2022(Acc.)*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 04 & 05 No. Question in are most important for the Examinee of 2025 ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2022 No-08**

Padma Ltd. is currently running at 100% capacity and produces 8,000 units, cost per unit as per details below :

	Taka
Materials	17
Labour	9
Factory overheads	6 (Tk 4 variable)
Administrative overheads	3 (Fixed)

The current selling price is Tk. 45 per unit. Prepare flexible budget for 80% and 100% capacity.

Solution : হিসাববিজ্ঞান ইজিপ্রেপ (৩য় খণ্ড) বইয়ের ২৯২ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–14**

The expenses budget for the production of 10,000 units in a factory are as follows :

Particulars	Pet unit
Raw material	Tk. 90.00
Labour	Tk. 25.00
Variable overhead	Tk. 20.00
Fixed overhead (Tk. 1,00,000)	Tk. 10.00
Variable expense (Direct)	Tk. 5.00
Selling expenses (10% fixed)	Tk. 15.00
Distribution expense (20% fixed)	Tk. 10.00
Administrative expense (Tk. 60,000)	Tk. 6.00

You are required to prepare a flexible budget for the production of 5,000 units and 8,000 units.

Solution : ২৯৫ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 15**

From the following data prepare a flexible budget of Jamuna Ltd. producing a single product for 80% and 100% capacity level. The break up of existing unit cost at 70% working level is :

Raw materials	Tk. 8
Wages	Tk. 4
Variable overhead	Tk. 10

Besides the above, the semi-variable costs for producing 14,000 units are :

Indirect materials	Tk. 47,000
Indirect labour	Tk. 31,200
Repairs	Tk. 24,000

The variable costs per unit included in semi-variable costs are indirect materials Tk. 1.00, labour Tk. 1.50 and repairs Tk. 1.25.

The fixed costs are :

Factory overhead	Tk. 26,000
Administrative overhead	Tk. 46,000
Selling overhead	Tk. 42,000 (Tk. 1.00 per unit variable)
Present output:	14,000 units
Selling price per unit	Tk. 38.

Solution : ২৯৬ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2016 No-14; 2017 No-15**

On the basis of standard working level (100% normal capacity), the following information is given about Nitol Industry :

Standard working level (100%) = 4,000 labour hours.

Account titles	Fixed expenses Tk.	Variable expenses per unit Tk.
Insurance	5,000	
Taxes	4,000	
Depreciation	8,000	
Supervisor salaries	12,000	
Variable expenses :		
Indirect materials and supplies		3.00
Indirect wages		2.00
Miscellaneous		0.50
Semi-variable expenses :		
Maintenances	5,000	1.00
Power house	8,000	0.50
General	3,000	0.50

Required : Prepare a flexible budget at 90%, 100% and 110% working level of indirect production expenses and calculate expense rate at these level.

Solution : ২৯৮ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–14**

A factory is currently working at 50% capacity and produces 20,000 units at a cost of Tk. 200 per unit as below :

Materials	Tk. 80
Labour	Tk. 50
Factory overhead (50% fixed)	Tk. 40
Administrative overhead (40% fixed)	Tk. 30

The selling price is Tk. 225 per unit. At 60% capacity level material cost per unit increases by 5% and selling price per unit falls by 2%. At 80% capacity level, materials cost per unit increases by 10% and selling price per falls by 4%.

Required :

Prepare a flexible budget at 50%, 60% and 80% capacity level.

Solution : ৩০০ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2019 No–14**

Hafiz Ltd. has been operating at 60% capacity (12,000 units). The following information are obtain from the records of the company for the year 2018:

Direct material	1,50,000
Direct wages	80,000
Direct expenses	42,000
Works overhead (60% variable)	26,000
Administrative overhead (fixed)	48,000
Selling overhead (40% variable)	34,000
Sales	4,80,000

The boost up sales above 75% level of activity, a reduction of 5% in the selling price is contemplated. On the basis of above information, you are required to prepare a flexible budget at 50%, 75% and 90% capacity level.

Solution : ৩০১ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. 2018 No-16**

From the following details against production of 8,000 units in 2016, Prepare flexible budget for 10,000 and 12,000 units for 2017 :

Accounts titles	Taka
Materials	40,000
Direct expenses	16,000
Wages	30,000
Factory expenses	20,000
Administrative expenses	40,000
Selling expenses	30,000
Total expenses	1,76,000
Sales	2,40,000
Profit	Tk. 64,000

Materials cost is likely to go up by 2% per unit. Wages are paid at a piece rate of Tk. 3 per unit. The fixed benefits paid to worker are likely to go up by 10%. Administrative expenses may go up by 5% due to normal increment. Selling expenses rise by 5% against increase of 20% or any part there of above 8,000 units. All selling prices are to be reduced by 10% if production above 10,000 units are to be sold.

Solution : ৩০২ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2013 No-15(b); 3rd (Finance) 2017 No-09; 2019 No-16; MBA Prell (Mgt.) 2014 No-4(b)**

A factory is currently running at 50% capacity and produces 10,000 units at a cost of Tk. 180 per unit as per details below :

Material Tk. 100, wages Tk. 30, factory overhead Tk. 30 (40% fixed), administration overheads Tk. 20 (50% fixed). The current selling price is Tk. 200 per unit.

At 60% capacity working, material cost per unit increased by 2% and selling price per unit falls by 2%. At 80% capacity working material cost per unit increased by 5% and selling price per unit falls by 5%.

Prepare a marginal cost statement showing the total cost and profit for the three capacity levels. Also give your comments on the profitability at these levels of performance.

Solution : ৩০৫ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 06**

The monthly budgets for the manufacturing overhead of a concern for two levels of activity are as follows :

Capacity	60%	100%
Budgeted production units	600	1,000
Cost items:	Taka	Taka
Indirect wages	12,000	20,000
Materials	9,000	15,000
Maintenance	11,000	15,000
Power and fuel	16,000	20,000

Required : Segregate variable costs and fixed costs of the items by using high and low points method.

Solution : ৩০৮ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's)Old-2014 No-12, Mgt-2017 No-13; 2022 No–11; (MBA Preli.) (Management) 2018 No–13 ,**

Following are the monthly budget for the manufacturing overhead of a cor for the two level of activity :

Capacity	60%	100%
Budgeted production (units)	1,200	2,000
Indirect wages	Tk. 1,200	Tk. 2,000
Consumable stores	Tk. 900	Tk. 1,500
Maintenance	Tk. 1,100	Tk. 1,500
Power and fuel	Tk. 1,600	Tk. 2,000
Depreciation	Tk. 6,000	Tk. 6,000
Insurance	Tk. 2,000	Tk. 2,000

Required :

(i) Indicate which of the items are fixed, variable and semi-variable.

(ii) Prepare a flexible budget for 70%, 80% and 90% activity level.

Solution : ৩১৩ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–16**

The following are the monthly budget for the manufacturing overhead of a concern for two levels of activity :

Capacity	60%	100%
Budgeted production (units)	600	1,000
Indirect wages	Tk. 1,20,000	Tk. 2,00,000
Consumable store	90,000	1,50,000
Maintenance	1,10,000	1,50,000
Power and fuel	1,60,000	2,00,000
Depreciation	6,00,000	6,00,000
Insurance	2,00,000	2,00,000

Required :

(a) Indicate which of the items are fixed, variable and mix.

(b) Prepare a flexible budget for 70%, 80% and 90% activity level.

Solution : ৬৪৪ পৃষ্ঠার **Question No-16** নং প্রশ্নের উত্তর দেখুন।

Chapter 09 Standard Costing

(প্রমাণ ব্যয় হিসাব)

Part-A : Brief Questions & Solutions :

ooo 01. **What is normal standard?**

Ans : Normal standard is the average standard that is anticipated to be attainable over a future period long enough to cover one trade cycle.

oooo 02 **What is standard cost?** 99%

[Test Exam : B.M, Edward, Uttara town & Eden College; N.U. 2012, 2013, 2015 (Old); 2019, 2021 (Acc.); 2020 (Finance); 2021 (Mgt.)]

Or, Define standard cost.

Or, What do you mean by standard costing?

Ans : Standard costing is technique of costing which uses standards for cost and revenue for the purpose of control through variance analysis.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo 03. **What is current standard?**

Ans : Current standard is a standard established for use over a short period of time related to current period of using standard.

ooo 04. **How many types of standard?**

Ans : Standard may be divided into the following five classes: (i) Basic Standard; (ii) Ideal Standard; (iii) Expected Standard; (iv) Normal Standard; (v) Current standard;

ooo 05. **What do you mean by 'Labour Cost Variance'?** 99% *[N.U. 2017 (Finance)]*

Ans. Labour cost variance is the difference between standard direct labour cost and actual direct labour cost incurred for the production achieved.

ooo 06. **What is favorable variance?**

Ans : Any variance that has favourable effect on profit is called favourable variance.

ooo 07. **What is unfavourable variance?**

Ans : Any variance that has an adverse effect on profit is called unfavourable variance.

ooo 08. **What is uncontrollable variance?**

Ans : An uncontrollable variance is one which is not amenable to control by particular individual of departmental head. This variance arises due to factory beyond the control of management.

ooo 09. **How many types of variance?** 99%

Ans : The following three classes of variance: (i) Cost Variance; (ii) Sales Variance; (iii) Profit Variance.

ooo 10. **How many types of total cost variance?**

Ans : Total cost variance is divided into variance for three elements of cost, namely, (i) Material Cost Variance; (ii) Labour Cost Variance; (iii) Overhead Variance.

ooo 11. **What is material cost variance?**

Ans : Material cost variance is the different between the standard direct material cost of the actual production volume and the actual cost of direct material.

oooo 12 **What is the material price variance?** 99%

[N.U. 2017 (Mgt.); 2020 (Finance)]

Ans : It refers to that portion of material cost variance which is due to the different between the standard prices specified and the actual price for direct materials used.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 13. **What is material usage variance?**

Ans : The material usage variance is the difference between actual usage and expected usage. It measures the extent of utilization of materials in production.

oo 14. **What is material mix variance?**

Ans : Material mix variance is the difference between total quantity of materials in standard proportion, priced at the standard price and the actual quantity of materials used, also priced at standard price.

ooo 15. **What is material yield variance?**

Ans : It is the differenced between standard yield of the actual material input and the actual yield, both valued at the standard material cost.

ooo 16. **What is labour cost variance?** 99%

Ans : Labour cost variance is the difference between standard direct labour cost and actual direct labour cost incurred for the production achieved.

oo 17. **What is labor rate variance?**

Ans : Labour rate variance is the difference between the standard and the actual direct labor rate per hour for the total hours worked.

18. *What is labor efficiency variance? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Acc.)]*

Ans : Labour efficiency variance is the difference between the standard hours for the actual production achieved and the hours actually worked, valued at the standard labour rate.

ooo 19. **What is labour mix variance?**

Ans : It refers to that portion of labour efficiency variance which is due to the difference between the standard grades of workers specified and actual grades of workers utilized.

ooo 20. **What is idle time variance?** 99%

Ans : It is that portion of labour efficiency variance which is due to abnormal circumstances like strike, lock-outs, power failures, machine breakdown etc. *[N.U. 2018 (Acc.)]*

oo 21. **What is overhead variance?**

Ans : Overhead variance is the difference between the standard cost of overhead absorbed in the output achieved and the actual overhead incurred. *[N.U. 2012 (Old); 2019 (Mgt.)]*

ooo 22. **What is fixed overhead cost variance?**

Ans : It represents the difference between the standard fixed overhead cost charged to production and actual fixed overhead cost incurred.

ooo (23) **What is variable overhead spending variance?** 99% *N. U. 2023 (Acc.)]*

Ans : Variable overhead spending variance is the differences between the actual variable overhead cost incurred during a period and the standard cost that should have been incurred based on the actual activity of the period.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 24. **Why standard costing is important?**

Ans : Standard costing is important because, standard costing increases institutional working efficiency, determine depreciation and losses and control cost of the business.

oo 25. **What is the main characterizing of standard costing?**

Ans : All elements of cost to be pre-determined. Such as materials, labour and overhead costs are pre-determined.

ooo 26. **What is basic (fixed or static) standard?**

Ans : This standard remains constant for a long period of time and hence it is also turned as long-term standard. This standard may be suitable for those items of cost which are fixed in nature (such as managerial salaries, interest on capital, etc.).

oo 27. **What is historical standard?**

Ans : The capacity to be utilized on the basis of past capacity, present situation and future changes is called expected capacity. A standard which is set up on expected capacity is called expected standard.

ooo 28. **How can you set a standard?**

Ans : I can set a standard on the basis of normal activity level of same nature business. It must be verified by the specialist of related subject.

ooo 29. **What is the main advantage of standard costing?** 99% *[N.U. 2019 (Finance)]*

Ans : The main advantage of standard costing is to turn out disadvantage of historical costing.

oo 30. **What do mean by historical cost?**

Ans : Historical cost that type of cost which has been expended and no expectation to recover this cost.

ooo 31. **What is expected (attainable or practical) standard?**

Ans : Expected standard is a standard which can be attained if a unit of work is carried out efficiently, a machine properly operated, materials properly used. This standard is more realistic, in nature and is best suited from cost contral point of view.

oo 32. **What is controllable variances?**

Ans : Controllable variance is one which is amenable to control by a particular individual or departmental head who is responsible for the variance. The management should place more emphasis on this variance as corrective actions may be taken to control this variance.

ooo 33. **What is high low method?** *[N.U. 2017 (Acc.)]*

Ans. High low method such method which to calculate variable overhead rate.

ooo (34) **What is variance?** 99% *[N.U. 2020, 2022 (Acc.)]*

Ans. According to CIMA variance analysis is a process of determining variance and causes of its.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত 99% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 35. **What types of organization makes use of standard costing?** *[N.U. 2019 (Mgt.)]*

Ans : The organization those are relating to production use standard costing.

ooo 36. **Give two example of non-cash expenses.** 99% *[N.U. 2016 (Finance)]*

Ans : (i) Depreciation and (ii) Written of goodwill.

ooo 37. **Write down the formula of segregation semi-variable cost by high-low method.** 99% *[N.U. 2016 (Finance)]*

Ans : High low method $= \frac{\text{High value} - \text{Low value}}{\text{High unit} - \text{Low unit}}$

Part- B & C (Short & Broad Questions):

০০ 01. (a) What is standard cost? [Ans হিসাববিজ্ঞান ইন্টেলেক্ট (৩য় খণ্ড) বইয়ের 322 Page 9.14 No.]
Or, What do you mean by standard cost?
(b) Discuss the merits and limitations of standard costing. [Ans : 326 Page 9.29 No.]
[Test Exam : Suhrawardy, Comilla, Haraganga, Sirajgonj, Tong, Sachal, Rajandra College; N.U (Acc)-2015]

০০০ 02. **Discuss the advantages of standard cost? What are the potential problems with the use of standard costs? 99% [Ans : 319+324 Page 9.04+9.21 No.]** *[Test Exam : Edwar, Bagbodna, B.M., Suhrawardy & Shahid Munsur Ali College; N.U-2013; (Acc.)-2016, 2017; (Finance)-2017]*

০০০ 03. **Discuss the method of standard costing. 99% [Ans : 319 Page 9.02 No.]**
[N.U. 2013, (Acc)-2015; (Mgt.)-2017; (Finance)-2018]
Or Discuss the methods for determining standards for standard costing.

০০০ (04) **"Standard costing is the tools for effective control of cost." –Explain the statement. 99%**
[Ans : 321 Page 9.12 No.] *[N.U. 2019, 2023 (Acc.)]*
Or, Why variance analysis is requlred?

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 02 & 04 No. Questions in are most important for the Examinee of 2025 ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

০০০ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2019 No– 08; (Finance) 2023 No– 08**
A budget is prepared for producing 1,000 units of product. The cost of 1,100 units of raw materials was fixed at Tk. 1,320 @ of Tk. 1.20 per unit. The actual cost of 1,120 units of raw materials @ of Tk. 1.25 per unit was Tk. 1,400.
Calculate : (i) Material usage variance; (ii) Materials price variance.

Solution : হিসাববিজ্ঞান ইন্টেলেক্ট (৩য় খণ্ড) বইয়ের ৩৩১ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.-2016 No-17(c)**
A Chemical mixture PX is prepared by mixing chemical? P and Chemical X in the proportion of 1 : 5. The standard prices of these chemicals are Tk. 5 and Tk. 0.50 per kg respectively. During the last production run 7,140 kg of P were mixed with 40,320 kg of X.
Compute the material mix variance.

Solution : ৩৩১ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Mgt. 2017 No-09; (Finance) 2020 No-06; 2022 No– 09**
The following are the particulars regarding the standard and the actual production of the product X.

Standard quantity of material per unit	5 kg
Standard price	Tk. 5 per kg.
Actual number of units produced	400
Actual quantity of materials used	2,200 kg
Price of material	Tk. 4.80 per kg.

Calculate all variances.

Solution : ৩৩১ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2019 No– 15**

Mahin Ltd. has adopted standard costing furnishes the following intonation :

Standard data :

100 kg materials for finished product of 70 kg. The price of materials Tk. 10 per kg.

Actual data :

Output	2,20,000kg
Materials used	2,80,000 kg
Cost of materials	Tk. 25,20,000

Required :

(i) Materials cost variance
(ii) Materials price variance
(iii) Materials usage variance

Solution : ৩৩৩ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. (MBA Preli.) (Mgt.) 2014 No- 5(b); (Acc.) 2014 No- 8(c); 2017 No- 6(c); 3rd (Old) 2012 No- 17; (Mgt.) 2017 No- 15; 2021 No–17**

The standard mix for producing 8,000 bottles of a product F is :

Material A : 1,000 litters @ Tk. 0.10
Material B : 2,000 litters @ Tk. 0.40

During June, 10,000 bottles of the product F were produced from an input of :

Material A : 1,500 litters @ Tk. 0.11
Material B : 3,300 litters @ Tk. 0.37

Required :

Compute the material price variance, material mix variance, material usage variance and material yield variance.

Solution : ৩৩৪ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০০ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2020 No– 17**

The standard mix for producing 8,000 bottles of product-M is :

Material A : 1,000 liters @ Tk. 0.20
Material B : 2,000 liters @ Tk. 0.80

During March, 2019, 10,000 bottles of product-M were produced from input of—

Material A : 1,500 liters @ Tk. 0.25
Material B : 3,300 liters @ Tk. 0.75

Required : Compute the material price, usage, mix, and yield variance for March.

Solution : ৩৩৭ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2024 No–14**

The standard cost and actual cost of materials per unit are given below :

Standard cost :

Material A : 5 kg @ Tk. 8
Material B : 10 kg @ Tk. 6

Actual cost :

Material A : 8 kg @ Tk. 7
Material B : 8.5 kg @ Tk. 7

During a week, 400 units are produced calculate :

(i) Materials price variance
(ii) Material mix variance
(iii) Material usage variance
(iv) Material cost variance
(v) Material yield variance

Solution : ৬৪২ পৃষ্ঠার **Question No-14** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.- 2021 No–17**

The standard cost per unit :

Material A :	5kg @Tk. 10	Tk. 50
Material B :	7kg @Tk. 15	Tk. 105
	Total	Tk. 155
Actual cost for 100 units :		
Materials A :	600kg @ Tk. 9	Tk. 5,400
Materials B :	650kg @ Tk. 16	Tk.10,400
	Total	Tk. 15,800

Required :

(a) Material price variance
(b) Material usage variance
(c) Material mix variance
(d) Material yield variance
(e) Material cost variance.

Solution : ৩৪৪ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2017 No-16; 2023 No–17**

Standard material for 100 units are given below :

Material P	45 units @ Tk. 2 each	Tk. 90
Material Q	40 units @ Tk. 4 each	Tk. 160
Material R	25 units @ Tk. 6 each	Tk. 150
	110	Tk. 400
Standard loss	(10)	
	100	

Actual production 2,000 units and actual material usages is as follows :

Material P	1,000 units @ Tk. 1.90	= Tk. 1,900
Material Q	850 units @ Tk. 4.20	= Tk. 3,570
Material R	450 units @ Tk. 6.50	= Tk. 2,925
	2,300 units	= Tk. 8,395

Calculate material cost variance, material price variance, material mix variance, material yield variance and material usage variance.

Solution : ৩৪৬ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.-2017 No-05; 2022 No-09**

From the following direct labour specification of Piyal Limited/Afia Ltd. you are required to calculate labour variances

Standard direct labour hours 20

Standard direct labour rate Tk. 2.50.

During a period 250 units were produced and direct labour amounted to Tk. 15,000 for 5,600 hours.

Solution : ৩৫১ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2023 No–11(b)**

From the following information, calculate labour variance.

Standard cost per unit :

10 hours Tk. 90 per hour

Actual cost for producing 200 units :

2,100 hours @ Tk. 80 per hour = Tk. 1,68,000.

50 hours are lost due to abnormal losses.

Solution : ৩৫১ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc.-2018 No-17**

A manufacturing company produces plastic toys and standard costing system is in operation. The following data is available for December, 2017 :

(i) Standard cost of production (per toy)

Direct materials 0.25 meter @ Tk. 2	= 0.50
Direct wages 0.20 hours @ Tk. 1.70	= 0.34
Production overhead	= 0.80
Total production cost	1.64

(ii) Budgeted production : 17,500 toys.

(iii) All production overhead is fixed.

(iv) Actual data for the period is as follows :

(a) Production : 15,000 toys

(b) Direct materials used : 3,900 meters @ Tk. 1.80 per meter.

(c) Direct wages paid : 2,850 hours @ Tk. 1.90 per hour.

(d) Production overhead incurred Tk. 14,740.

Calculate material and labour variances.

Solution : ৩৫৪ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

BP BANGLADESH PUBLICATION

গ্রন্থ সহায়ক : উপরিউক্ত সাজেশন্সের পূর্ণাঙ্গ উত্তরপত্র পেতে.....

ব্যতিক্রম হিসাববিজ্ঞান ইজিপ্লাস (৩য় খণ্ড) বইটি সংগ্রহ করুন।

জাতীয় বিশ্ববিদ্যালয়ের প্রশ্নাবলি ও উত্তরপত্র

Management Accounting (In English); Subject Code : 232507

National University

[BBA Hon's 3rd year; Exam-2019 (Held on-12/02/2020)]

(Department of Accounting)

Subject : Management Accounting (In English)

Subject Code : 232507

Time : 4 hours; **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) Elaborate "ICMAB".

Ans : The elaboration of ICMAB is Institute of Cost and Management Accountants of Bangladesh.

(b) What do you mean by ethics?

Ans : The standards of conduct by which ones action are judged as right of wrong, honest or dishonest, fair or not fair.

(c) What is sunk cost?

Ans : Sunk cost is the cost that has already been incurred and that cannot be changed by any decision made now or in the future.

(d) What is opportunity cost?

Ans : The potential benefit that is lost or sacrificed when the selection of one course of action makes it necessary to give up a competing course of action is called opportunity cost.

(e) What is conversion cost?

Ans : Conversion cost is the combined form of direct labour and manufacturing overhead.

(f) What is absorption costing?

Ans : A costing method that includes all manufacturing cost direct materials, direct labour and both variable and fixed manufacturing overhead in the cost of a unit of product. Absorption costing is also referred to as the full cost method.

(g) What is margin of safety?

Ans : The excess of budgeted (or actual) sales over the break-even volume is margin of safety.

(h) What is contribution margin (CM)?

Ans : The contribution margin per unit expressed as a percentage of selling price per unit. This term is synonymous with profit-volume ratio.

(i) What is joint products?

Ans : a joint product is a product that results jointly with other products from processing a common input.

(j) What is relevant cost?

Ans : Relevant cost are those expected future cost that differ among alternative course of action.

(k) What is operating budget?

Ans : The operating budget focuses on the income statements and its supporting schedules or in an organization with no sales revenues on budgeted expenses and supporting schedules.

(l) What is standard costing?

Ans : Standard costing is technique of costing which uses standards for cost and revenue for the purpose of control through variance analysis.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Discuss the importance of ethics in business.

[Ans : ব্যাতিক্রম হিসাববিজ্ঞান হ্যান্ডবুক (৩য় খণ্ড) বইয়ের **Page No. 7; Ques. No. 1.09]**

3. What factors will you consider at the time of "Make or Buy" decision? ***[Ans : Page No. 14 ; Ques. No. 1.29]***

4. Show the differences between product cost and period cost. ***[Ans : Page No. 21 ; Ques. No. 2.17]***

5. Sales Tk. 2,00,000, P/V ratio 25%, Fixed cost Tk. 80,000.
Required : (i) BEP (ii) Margin of safety

Solution : ***Page No-147; Problem No-07***

6. The following data are related to the Akter Company Ltd . :

	Ending	Inventories Beginning
	Taka	Taka
Finished goods	95,000	1,10,000
.Work-in-process	60,000	50,000
Direct materials	85,000	80,000

Cost incurred during the period :	Taka
Cost of goods available for sales	6,84,000
Total manufacturing costs	5,84,000
Factory overhead	1,67,000
Direct materials used	1,90,000

Required :

(i) Direct Labour cost

(ii) Direct materials purchased during the period.

Solution : ***Page No-34; Problem No-08***

7. Saiam Ltd. is examining the profitability of investing in new project. The following information is available in respect of a project :

Investment	Tk. 10,00,000
Salvage value	Tk. 60,000
Life of the project	20 years
Annual profit of the project (before tax)	1,80,000

Tax rate 40%.

Required : (i) PBP (ii) ARR.

Solution : ***Page No-258; Problem No-41***

8. A budget is prepared for producing 1,000 units of product. The cost of 1,100 units of raw materials was fixed at Tk. 1,320 @ of Tk. 1.20 per unit. The actual cost of 1,120 units of raw materials @ of Tk. 1.25 per unit was Tk. 1,400.

Calculate :

(i) Material usage variance

(ii) Materials price variance.

Solution : ***Page No-331; Problem No-01***

9. Show the difference between fixed and variable cost.

[Ans : Page No. 19 ; Ques. No. 2.09]

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. Discuss the various techniques of Management Accounting. ***[Ans : Page No. 7; Ques. No. 1.10]***

11. The following data is available for the month of July, 2018 of Sakib Ltd. :

Beginning Inventory	Nil
Sales units	8,000
Production units	10,000
Selling price (per unit)	Tk. 40
Variable manufacturing cost (per unit)	Tk. 12
Fixed factory overhead (per unit)	Tk. 5
Fixed factory overhead (total)	Tk. 35,000
Fixed commercial expense	Tk. 18,000

(i) Prepare income statement under variable costing and absorption costing.

(ii) Prepare a reconciliation statement.

Solution : ***Page No-115; Problem No-36***

12. Tania Ltd. using 50% of capacity produces 30,000 units last year for market. Of the total cost Tk. 90,000 was variable cost and Tk. 70,000 was fixed cost. Due to trade depression, the company has to reduce the price from Tk. 8 to 6 per unit during the current year. The company has also decide to increase production to 50,000 units and to sell the excess production to foreign market at Tk. 5.00 per unit. Selling expenses for the foreign sales will be Tk. 0.50 per unit. Fixed cost will increase by Tk. 5,000.

Required :

(i) Profit or loss at home and foreign market during the current year.

(ii) Break-even sales during the current year.

Solution : ***Page No-177; Problem No-48***

13. Salam Ltd. has the following forecasted data :

Details	September	October
	Taka	Taka
Cash Sales	70,000	80,000
Credit Sales	90,000	80,000
Cash Purchase	30,000	25,000
Credit purchase	75,000	65,000
Purchase discount	8,000	4,000
Accounts Payable (Beginning)	12,000	15,000
Accounts Payable (Ending)	14,000	11,000
Cash Operating Expenses	48,000	13,000

Net sales on credit are collected 50% in the month of sales, 40% in the following and 10% in the second following month. Such sales in July and August were Tk. 1,20,000 and 1,40,000 respectively. The estimated September 1, Cash is Tk. 16,000.

Prepare a cash budget for the month of September and October.

Solution : ***Page No-241; Problem No-23***

14. Hafiz Ltd. has been operating at 60% capacity (12,000 units). The following information are obtain from the records of the company for the year 2018:

Direct Material	1,50,000
Direct wages	80,000
Direct expenses	42,000
Works Overhead (60%.variable)	26,000
Administrative Overhead (fixed)	48,000
Selling Overhead (40% variable)	34,000
Sales	4,80,000

The boost up sales above 75% level of activity, a reduction of 5% in the selling price is contemplated. On the basis of above information, you are required to prepare a flexible budget at 50%, 75% and 90% capacity level.

Solution : ***Page No-301; Problem No-12***

15. Mahin Ltd. has adopted standard costing furnishes the following intonation :

Standard data :

100 kg materials for finished product of 70 kg. The price of materials Tk. 10 per kg.

Actual data :

Output	2,20,000kg
Materials used	2,80,000 kg
Cost of materials	Tk. 25,20,000

Required :

(i) Materials cost variance

(ii) Materials price variance

(iii) Materials usage variance

Solution : ***Page No-333; Problem No-05***

16. (a) Discuss the advantages of capital budgeting.
[Ans : Page No. == ; Ques. No. ==]

(b) Crown Ltd. is considering a proposal to install a new machine. The project will cost Tk. 1,00,000 and its expected life is five years. The estimated cash-inflows before tax (CFBT) of the project are as follows :

Year	1	2	3	4	5
CFBT	25,000	30,000	35,000	50,000	40,000

The tax rate is 50% and charges depreciation on straight line basis :

Required :

(i) PBP; (ii) ARR; (iii) NPV (10% discount rate).

Solution : *Page No-268; Problem No-51*

17. (a) Show the differences between Financial Accounting and Management Accounting.
[Ans : Page No. 4 ; Ques. No. 1.13]

(b) Why variance analysis is required?
[Ans : Page No. 321 ; Ques. No. 9.12]

National University

[BBA Hon's 3rd year; Exam-2020 (Held on-16/03/2022)]

(Department of Accounting)

Subject : Management Accounting (In English)

Subject Code : 232507

Time : 4 hours; **Full marks :** 80

[**N.B.** Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) **What is management accounting?**

Ans : Management accounting is the process of identification measurement, accumulation, analysis, preparation, interpretation and communication of information that assists executives in fulfilling organizational objectives.

(b) **What is prime cost?**

Ans : Prime cost is the combined form of direct materials, direct labour and direct expenses.

(c) **What are the elements of cost?**

Ans : The elements of cost are : 1. Material, 2. Labour and 3. Overhead.

(d) **What is differential cost?**

Ans : A differential cost between two alternatives is called differential cost.

(e) **What is activity based costing?**

Ans : Activity based-costing refers a costing system by identifying individual activities as the fundamental cost objects.

(f) **Write down the formula of profit volume ratio.**

Ans : $\text{P/V ratio} = \dfrac{\text{Contribution per unit}}{\text{Selling price per unit}}$

(g) **What is value chain?**

Ans : Value chain is the set of business functions that add value to the products of services of an organization.

(h) **What is product cost?**

Ans : All costs that are involved in the purchase or manufacture of goods. In case of manufactured goods these costs consist of direct materials, direct labour and manufacturing overhead. Also see inventor able costs.

(i) **What do you mean by flexible budget?**

Ans : A budget which is designed to changed in accordance with the level of activity actually attained is called flexible budget.

(j) **What is capital rationing?**

Ans : Capital rationing refers to the choice of investment proposals under financial constraints in terms of given size of capital.

(k) **What is cash budget?**

Ans : Cash budget is plan indicating expected inflows (receipts) and outflows (disbursements) of cash.

(l) **What is variance?**

Ans : Accounting to CIMA variance analysis is a process of deterring variance and causes of its.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Discuss the objectives of management accounting.
[Ans : ব্যাচেলর হিসাববিজ্ঞান বইয়ের (৩য় খণ্ড) বইয়ের **Page No. 5; Ques. No. 1.04]**

3. Show the differences between variable costing and absorption costing. *[Ans : Page No. 75 ; Ques. No. 3.14]*

4. Sales Tk. 12,00,000, CM ratio 40%, and MS ratio 25%.
Required : (a) Variable cost in Taka, (b) Profit in Taka. **Solution :** *Page No-147; Problem No-09*

5. Amazon Company produces radios for cars. The following cost information is available for the period ended December 31, 2019 :
Materials put into production Tk. 1,20,000, of which Tk. 80,000 was for direct materials. Factory labour costs for the period Tk. 90,000, of which Tk. 25,000 was for indirect labour. Factory overhead costs for utilities Tk. 40,000. Work-in-process beginning and ending value were Tk. 12,000 and Tk. 20,000 respectively.
Required : Calculate cost of goods manufactured (show prime costs and total manufacturing costs in your calculations).
Solution : *Page No-34; Problem No-07*

6. The monthly budgets for the manufacturing overhead of a concern for two levels of activity are as follows :

Capacity	60%	100%
Budgeted production units	600	1,000
Cost items:	Taka	Taka
Indirect wages	12,000	20,000
Materials	9,000	15,000
Maintenance	11,000	15,000
Power and fuel	16,000	20,000

Required : Segregate variable costs and fixed costs of the items by using high and low points method.

Solution : ***Page No-308; Problem No-18***

7. Meghna Ltd. is considering to purchase a new machine. The machine will cost Tk. 3,00,000. The cash inflows per year would be Tk. 80,000. The machines economic life is 10 years and cost of capital is 12%. The tax rate of the company is 45%. Calculate NPV and suggest whether the machine should be purchased or not. Assume PVIFA for Tk. 1.00 @ 12%, 10 years is 5.6502.

Solution : ***Page No-257; Problem No-39***

8. Delta Company is considering expansion of its plant capacity. Fixed costs amount to Tk. 4,20,000 and are expected to increase by Tk. 1,25,000 when plant expansion is completed. The present plant capacity is 80,000 units per year. Capacity will increase by 50% with the expansion. Variable costs are currently Tk. 6 per unit and are expected to go down by Tk. 1 per unit with the expansion. The current selling price is Tk. 20 per unit and expected to remain same under either alternative. Which alternative is better expand or not to expand? Why?

Solution : ***Page No-193; Problem No-02***

9. Gold Star Ltd. has budgeted sales of its products for the next four months as follows :

Month	Sales in units
April	50,000
May	75,000
June	90,000
July	80,000

The company is now in the process of preparing a production budget for the second quarter. Past experience has shown that end-of-month inventory levels must equal 10% of the following month's sales. The inventory at the end of March was 5,000 units.

Required : Prepare a production budget for the second quarter.

Solution : ***Page No-230; Problem No-08***

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. (a) Discuss the importance of ethics in business. ***[Ans : Page No. 7; Ques. No. 1.09]***

(b) Discuss the limitations of management accounting. ***[Ans : Page No. 6 ; Ques. No. 1.06]***

11. The Apex Company produce different types of leather goods. The sales revenue for the year 2020 Tk. 12,00,000. The cost related different information during the year were as follows :

Particulars	Taka
Raw material purchased	50,000
Freight in	2,000
Opening work-in-process	80,000
Opening finished goods	40,000
Opening raw materials	2,50,000
Closing work in process	70,000
Closing finished goods	60,000
Closing raw materials	60,000
Productive wages	1,70,000
Chargeable expenses	50,000
Factory overhead	2,50,000
Selling expenses	40,000
General and administrative expenses	10,000
Sale of scrap	5,000

Prepare a statement showing the following :

(i) Prime cost
(ii) Cost of goods manufactured
(iii) Cost of goods available for use
(iv) Cost of goods sold
(v) Cost of sales
(vi) Profit or loss.

Solution : ***Page No-41; Problem No-17***

12. The following particulars are available from the books of a manufacturing concern :

Normal capacity	20,000 units
Standard variable manufacturing cost per unit	Tk. 15
Variable marketing expenses per unit sold	Tk. 5
Fixed factory overhead	Tk. 40,000
Fixed marketing expenses	Tk. 25,000
Selling price per unit	Tk. 30

Selected data for the company's operations in the last year were as follows :

Beginning inventory	2,000 units
Sales	17,500 units
Production	18,000 units

Required : Prepare income statement using variable costing and absorption costing.

Solution : ***Page No-92; Problem No-20***

13. Otobi Ltd. manufactures and sells a single product. The company's sales and expenses for the last quarter were as follows :

	Taka
Sales (15,000 units)	9,00,000
Less : Variable expenses	3,60,000
Contribution margin (CM)	5,40,000
Less: Fixed expenses	4,32,000
Net operating income	1,08,000

Required :

(a) What is company's CM ratio?

(b) What is quarterly break even point in units sold and in sales taka.

(c) Compute company's margin of safety in taka and in percentage terms.

(d) How many units would have to be sold each quarter to earn a pre-tax profit of Tk. 1,80,000?

(e) If company can sell only 15,000 units but wanted to earn a pre-tax profit of Tk. 1,80,000 then what would be the selling price per unit?

Solution : ***Page No-175; Problem No-46***

14. A company making stock in the first quarter of the year is assisted by its bankers with cash facilities. The following are the relevant budget figures :

Month	Credit sales	Credit purchases	Wages and expenses
	(Taka)	(Taka)	(Taka)
November	1,20,000	83,000	10,000
December	1,28,000	96,000	10,000
January	1,80,000	1,40,000	11,000
February	1,16,000	1,20,000	10,000
March	84,000	40,000	12,000

Given the following further information you are required to prepare a cash budget for the quarter January to March. Other information are as follows :

(i) Budgeted cash at bank on January 1, Tk. 25,000.

(ii) Credit terms of sales are payable by the end of the month following the month of supply. On an average one-half of sales are paid on the due date, while the other half are paid during the next month. Creditors are paid during the month following the month of supply.

(iii) Wages and expenses are paid twice in the month of 1st and 16th respectively.

Solution : ***Page No-245; Problem No-26***

15. From the following data prepare a flexible budget of Jamuna Ltd. producing a single product for 80% and 100% capacity level. The break up of existing unit cost at 70% working level is :

Raw materials Tk. 8
Wages Tk. 4
Variable overhead Tk. 10
Besides the above, the semi-variable costs for producing 14,000 units are :
Indirect materials Tk. 47,000
Indirect labour Tk. 31,200
Repairs Tk. 24,000
The variable costs per unit included in semi-variable costs are indirect materials Tk. 1.00, labour Tk. 1.50 and repairs Tk. 1.25.
The fixed costs are :
Factory overhead Tk. 26,000
Administrative overhead Tk. 46,000
Selling overhead Tk. 42,000 (Tk. 1.00 per unit variable)
Present output: 14,000 units
Selling price per unit Tk. 38.

Solution : ***Page No-296; Problem No-06***

16. Sovon Ltd. manufactures 40,000 parts of specialized products. At this level of activity the cost per parts is as follows :

	Taka
Direct materials	5
Direct labour	10
Variable overhead	2
Fixed overhead	8
Total cost per parts	25

An outside supplier has offered to sell 40,000 parts each year to Sovon Ltd. for Tk. 22 per parts. If Sovon Ltd. accepts this offer, the facilities now being used to manufacture the parts could be rented to another company at an annual rental of Tk. 80,000. However, the company has determined that Tk. 6 per unit of the fixed overhead being applied to the parts would continue even the parts are purchased from outside.

Required : Should the parts be purchased from outside or continue to produce in the factory?

Solution : ***Page No-197; Problem No-09***

17. The standard mix for producing 8,000 bottles of product-M is :
Material A : 1,000 liters @ Tk. 0.20
Material B : 2,000 liters @ Tk. 0.80
During March, 2019, 10,000 bottles of product-M were produced from input of—
Material A : 1,500 liters @ Tk. 0.25
Material B : 3,300 liters @ Tk. 0.75
Required : Compute the material price, usage, mix, and yield variance for March.

Solution : ***Page No-337; Problem No-10***

National University

[BBA Hon's 3rd year; Exam-2021 (Held on-04/06/2023)]

(Department of Accounting)

Subject : Management Accounting (In English)

Subject Code : 232507

Time : 4 hours; **Full marks :** 80

[N.B. Answer Different questions of each group chronologically.]

Part-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is business ethics?

Ans : Business ethics refers to the standards of conduct or moral behavior as applied to the business practices.

(b) What are the elements of prime cost?

Ans : The elements of cost are: a. Material, b. Labour and c. Overhead.

(c) What is cost center?

Ans : Cost center is a segment of a company that has a control over generating revenue or the use of investment of funds.

(d) What is the elaboration of "ICAB"?

Ans : The elaboration of ICAB is "Institute of Chartered Accountants of Bangladesh."

(e) Define working capital.

Ans : Working capital means net working capital which is currents assets minus current liabilities.

(f) Write down any one application of C/M ratio.

Ans : Contribution margin ratio $= \frac{\text{Contribution}}{\text{Sales}}$.

(g) What is special order?

Ans : A one time order that is not considered part of the company's normal ongoing business is called special order.

(h) What is capital rationing?

Ans : Capital rationing refers to the choice of investment proposals under financial constraints in terms of given size of capital.

(i) What is standard cost?

Ans : Standard costing is technique of costing which uses standards for cost and revenue for the purpose of control through variance analysis.

(j) Define absorption costing.

Ans : A costing method that includes all manufacturing cost direct materials, direct labour and both variable and fixed manufacturing overhead in the cost of a unit of product. Absorption costing is also referred to as the full cost method.

(k) What is IRR?

Ans : IRR is the discount rate at which the present value of expected cash inflows from a project equals the present value of expected cash outflows of the project.

(l) Write down the formula of BEP.

Ans : BEP $= \frac{\text{Fixed cost}}{\text{Contribution per unit}}$

Part-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. What are the techniques of management accounting?

[Ans : বেস্ট কম্বিনেশন হিসাববিজ্ঞান গাইডবুক (৩য় খণ্ড) বইয়ের **Page No. 7; Ques. No. 1.10]**

3. Show the difference between product cost and period cost. ***[Ans : Page No. 21 ; Ques. No. 2.17]***

4. What are the advantages of flexible budget? ***[Ans : Page No. 287 ; Ques. No. 8.02]***

5. The following data relates to the 'X' Manufacturing Co. of a standard product during four weeks period of June 30, 2020.

Raw material consumed	Tk. 6,000
Wages	Tk. 9,000
Machine hours worked 1,000 hours	
Machine hour rate	Tk. 0.60
Office overhead: 25% on works cost	
Selling overhead per unit	Tk. 0.08

Units produced: 20,000 units

Units sold: 18,000 units @ Tk. 1.20 per unit.

You are required to prepare

(i) Cost of goods sold statement

(ii) An income statement.

Solution : ***Page No-52; Problem No-30***

6. Operation of 'Z' Company for the last year disclosed an M/S ratio 20% and a C/M ratio 50%. Fixed cost Tk. 40,000.

Required : (i) Break even sales (ii) The amount of profit (iii) The contribution margin.

Solution : ***Page No-150; Problem No-17***

7. 'Y' Manufacturing Ltd. is examining the profitability of invating in new project. The following information is avallable in respect of the project:

Investment	Tk. 12,70,000
Residual value at the end of life	Tk. 70,000
Life of the project	20 years
Annual profit of the project (before tax)	Tk. 1,80,000

Tax rate 50%

Required :

(i) Payback period

(ii) Net present value

(iii) Average rate of return.

Solution : ***Page No-260; Problem No-43***

8. ABC Company's sales budget for the third quarter of 2020 is given below :

Particular	July (Tk.)	August (Tk.)	Septembe r (Tk.)	Total (Tk.)
Budgeted sales	3,00,000	4,00,000	3,50,000	10,50,000

25 percent of a month's sales are collected in the month of sales, 65 percent is collected in the month following sales and remaining 10 percent is collected in the second month following sale. May sales totalled Tk. 2,50,000 and June sales totalled Tk. 2,50,000.

Required: Show the schedule of cash collections from sales by month and in total third quarter.

Solution : ***Page No-234; Problem No-14***

9. ABC Company currently makes all the components for a laser compact disk player. They have been approached by a company in Thailand to supply a main part for Tk. 50 each. Company's managers are convinced that the foreign firm can supply these parts with consistent quality and acceptable delivery schedules. An analysis of the cost records associated with this part shows the following :

Materials	Tk. 15
Labour (1 hour)	10
Overhead	40
	65

Normal volume is 10,000 units a month. Overhead consists of Tk. 15 per hour in costs that vary directly with labour hours, Tk. 5 per hour allocated for rent on machinery used exclusively to produce this part and Tk. 20 consisting of allocated general factory rent, supervision, accounting and computer costs.

Required : At 10,000 units per month, should company's managers to make or buy this component?

Part-C

[Answer any five questions.] $5 \times 10 = 50$

10. (a) Discuss the functions of management accounting.

[Ans : Page No. 11/ Ques. No. 1.22]

(b) Show the differences between management accounting and cost accounting.

[Ans : Page No. 9 / Ques. No. 1.15]

11 (a) Describe the importance of cash budget.

[Ans : Page No. 218 / Ques. No. 7.37]

(b) Distinguish between cash budget and cash flow statement.

[Ans : Page No. 218 / Ques. No. 7.39]

12. (a) Describe the various type of cost according to cost behaviour.

[Ans : Page No. 26 ; Ques. No. 2.30]

(b) Distinguish between zero based budget and traditional budget.

[Ans : Page No. 225 ; Ques. No. 7.72]

13. Tamanna Manufacturing Company uses a standard cost system in accounting for its product which it sells for Tk. per unit. The standard cost per unit is :

Direct materials	Tk. 7
Direct labour	Tk. 6
Variable factory overhead	Tk. 3
Fixed factory overhead (Based on normal capacity of 70,000 units)	Tk. 4
Total cost	Tk. 20

All variances are closed to cost of goods sold.

On June 1, 2020 there were 5,000 units of product on hand. During June 60,000 units were produced and 50,000 units were sold. Cost incurred during June were :

Direct materials	Tk. 4,10,000
Direct labour	Tk. 3,90,000
Variable factory overhead	Tk. 2,00,000
Fixed factory overhead	Tk. 2,50,000
Variable marketing overhead	Tk. 75,000
Fixed marketing overhead	Tk. 80,000

Required :

(a) Prepare income statement under variable costing.

(b) Prepare income statement under absorption costing.

Solution : ***Page No-103; Problem No-28***

14. A factory is currently working at 50% capacity and produces 20,000 units at a cost of Tk. 200 per unit as below :

Materials	Tk. 80
Labour	Tk. 50
Factory overhead (50% fixed)	Tk. 40
Administrative overhead (40% fixed)	Tk. 30

The selling price is Tk. 225 per unit. At 60% capacity level material cost per unit increases by 5% and selling price per unit falls by 2%. At 80% capacity level, materials cost per unit increases by 10% and selling price per falls by 4%.

Required :

Prepare a flexible budget at 50%, 60% and 80% capacity level.

Solution : ***Page No-300; Problem No-11***

15. The following data is extracted from the books of a company :

Annual sales	Tk. 2,00,000
Margin of safety ratio	30%
Profit volume ratio	40%

Required : Ascertain–

(a) Break-even in sales in taka
(b) Margin of safety in taka
(c) Annual profit
(d) Sales required to earn a profit of Tk 40,000
(e) Profit or loss on sale of Tk. 1,50,000.

Solution : ***Page No-154; Problem No-22***

16. A factory produces and supplies goods according to customers' specification. From the following particulars for the year 2020 prepare a statement of cost :

Inventory of raw materials (January 1, 2020)	Tk. 25,000
Inventory of finished goods (January 1, 2020)	Tk. 50,000
Inventory of raw materials (December 31, 2020)	Tk. 37,500
Inventory of finished goods (December 31, 2020)	Tk. 30,000
Purchase of raw materials	Tk. 1,62,500
Direct wages	Tk. 1,00,000
Factory overhead	Tk. 62,500
Office and administrative overhead	Tk. 31,250
Sales	Tk. 4,00,000

The factory is about to send a tender for supply of machine. The costing department estimates that the materials required would cost Tk. 60,000 and direct wages would cost Tk. 40,000. Factory overhead is charged at a percentage on prime cost and office and administrative overhead is charged at a percentage on factory overhead.

Compute the price of the tender to earn a profit of 20% on total cost.

Solution : ***Page No-64; Problem No-39***

17. The standard cost per unit :

Material A :	5kg @Tk. 10	Tk. 50
Material B :	7kg @Tk. 15	Tk. 105
	Total	Tk. 155
Actual cost for 100 units :		
Materials A :	600kg @ Tk. 9	Tk. 5,400
Materials B :	650kg @ Tk. 16	Tk.10,400
	Total	Tk. 15,800

Required :

(a) Material price variance
(b) Material usage variance
(c) Material mix variance
(d) Material yield variance
(e) Material cost variance.

Solution : ***Page No-344; Problem No-16***

National University

[BBA (Hon's) 3rd Year; Exam-2022 (Held in-06/05/2024)]

(Department of Accounting)

Subject Code : 232507

Subject Name : Management Accounting (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is cost sheet?

Ans : Cost sheet is a statement which is prepared usually to present the detailed costs of total output during a period at sequence.

(b) Define cost and expense.

Ans : Cost : Cost is a forgoing measured in monetary terms, incurred or potentially to be incurred to achieve a specific objective.

Expense : Expenses is a part of cost which is expired and deductible from revenue.

(c) What is relevant range?

Ans : The relevant range is the range of activity within which the assumption about variable and fixed cost behavior are effective.

(d) What is work in process?

Ans : Work in process consists of units of product that are only partially complete and will require further work before they are ready for sale to the customers.

(e) What is activity cost pool?

Ans : Set of costs which are incurred when certain operations are performed within the organization is called activity cost pool.

(f) What is margin of safety?

Ans : The excess of budgeted (or actual) sales over the break-even volume is margin of safety.

(g) What is decision making?

Ans : Decision making is choosing the best among the alternatives.

(h) What is opportunity cost?

Ans : The potential benefit that is lost or sacrificed when the selection of one course of action makes it necessary to give up a competing course of action is called opportunity cost.

(i) **What is master budget?**

Ans : The master budget is a set of interrelated budget, that constitute a plan of action for a specified time period.

(j) **What is relevant cash flow?**

Ans : The relevant cash flows is future incremental cash flow arising from the decision being made. This means that the cash flows are only relevant if they are future, incremental costs.

(k) **What is cost of capital?**

Ans : The appropriate interest rate is the cost of the finance provided to the company by shareholders and other investors, as investment projects have to earn a return at least sufficient to pay the interest on the funds they utilize.

(l) **What is variance?**

Ans : Accounting to CIMA variance analysis is a process of deterring variance and causes of its.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Describe briefly steps in budgetary control.

[Ans : [illegible] হিসাববিজ্ঞান ইন্টারপ্রেস (৩য় খণ্ড) বইয়ের Page No. 191; Ques. No. 6.20]

3. What are the characteristics of activity based costing? ***[Ans : Page No. 124 ; Ques. No. 4.02]***

4. What factors will you consider at the time of "Make or Buy" decision? ***[Ans : Page No. 14 ; Ques. No. 1.29]***

5. TAZ Ltd. costs of producing 2,500 units in January and 3,400 units in February were Tk. 21,830 and Tk. 24,980 respectively. The Co. plans to produce 2,000 units in March. How much would be the cost of production in March?

Solution : ***Page No-149; Problem No-15***

6. Tasin Ltd. has a profit of Tk. 35,000, fixed cost Tk. 40,160 and break even point Tk. 1,25,500. Find out the amount of sales.

Solution : ***Page No-145; Problem No-02***

7. Noor Co. has been producing and selling 30,000 units of its product per month with the following total costs:

	Taka
Direct materials	90,000
Direct labour	75,000
Variable factory overhead	60,000
Fixed factory overhead	80,000
Variable marketing and administrative expenses	45,000
Fixed marketing and administrative expenses	50,000

The normal sales price is Tk. 17 and plant capacity is 40,000 units. The company has received an offer from a special customer who would like to buy exactly 7,000 units of product for Tk. 9 per unit. Marketing and administrative expenses related to this special order would be Tk. 4,000.

Required : Should Noor Co. accept this special order?

Solution : ***Page No-195; Problem No-06***

8. Padma Ltd. is currently running at 100% capacity and produces 8,000 units, cost per unit as per details below :

	Taka
Materials	17
Labour	9
Factory overheads	6 (Tk 4 variable)
Administrative overheads	3 (Fixed)

The current selling price is Tk. 45 per unit. Prepare flexible budget for 80% and 100% capacity.

Solution : ***Page No-292; Problem No-01***

9. From the following direct labour specification of Afia Ltd. and you are required to calculate labour variances :

Standard direct labour hours 20

Standard direct labour rate Tk. 2.50

During a period 250 units were produced and direct labour amounted to Tk. 15,000 for 5,600 hours.

Solution : ***Page No-351; Problem No-21***

Group-C

[Answer any **five** questions.] $10 \times 5 = 50$

10. (a) How does management accounting assist the management of a business concern? ***[Ans : Page No. 4; Ques. No. 1.14]***

(b) Discuss the importance of ethics in management accounting. ***[Ans : Page No. 6 ; Ques. No. 1.07]***

11. (a) How does cost volume-profit analysis help in management decision making? ***[Ans : Page No. 140 ; Ques. No. 5.17]***

(b) "Variable cost is fixed per unit and fixed cost is variable in total"-Explain. ***[Ans : Page No. 29 ; Ques. No. 2.36]***

12. (a) What are the principal assumption underlying the break even analysis? ***[Ans : Page No. 139 ; Ques. No. 5.12]***

12. (b) Discuss the some common usage of flexible budget. ***[Ans : Page No. 289 ; Ques. No. 4.14]***

13. Asia Ltd. sells 1,200 units of a product at Tk. 92 per unit. Cost of direct materials is 30% of the cost of goods sold. The ratio of direct wages and factory overhead is 5 : 2. The administrative and selling expenses amount to Tk. 21,200. The net profit during the current year is Tk. 24,400.

The company plans to sell 1,500 units next year at Tk. 96 per unit. Wages rates and raw materials prices are expected to increase by 20% and 10% respectively. Factory overhead is applied on the basis of direct labour cost. Other cost will remain unchanged. Show the difference in net profit of the two years.

Solution : ***Page No-49; Problem No-25***

14. Tasnia traders sells its product for Tk. 30 per unit. The standard cost per unit is :

Account titles	Taka
Direct materials	10
Direct labour	7
Variable factory overheads	3
Fixed factory overheads (normal capacity of 20,000 units)	4
Total	24

At the beginning, there were 5,000 units of products on hand. During the year 15,000 units were produce and 15,000 units were sold. Per year selling expenses Tk. 20,000 plus 5% on sales.

Required :

(i) Prepare an income statement using variable costing

(ii). Prepare an income statement using absorption costing.

Solution : ***Page No-93; Problem No-21***

15. Joy company makes two products :

Product	Direct labour hours per unit	Production
P	0.80	10,000 units
Q	0.40	40,000 units

Additional information for the company :

(a) P requires Tk. 32 in direct materials per unit and Q requires Tk. 18.

(b) The direct rate is Tk. 15 per hour.

(c) P is more complex to manufacture than Q and they require special processing.

(d) The ABC system has the following activity cost pools:

Activity cost pool and activity measure	Estimated overhead cost (Tk.)	P	Q	Total
Machine set up (No. of set up)	72,000	100	300	400
Special processing (Machine houres)	2,00,000	5,000	–	5,000
General factory (Direct labour hours)	8,16,000	8,000	16,000	24,000

Required :

(i) Compute the activity rate for each activity cost pool.

(ii) Determine the unit cost of each product according to the ABC system including direct materials and direct labour.

Solution : ***Page No-132; Problem No-08***

16. The following data are obtained from the record of Jamuna Company :

	Taka	Taka
Sales (8,000 units @ Tk 25 each)		2,00,000
Direct materials consumed	80,000	
Variable overhead	25,000	
Labour	35,000	
Fixed overhead	36,000	1,76,000
Net profit		24,000

Required :

(i) The number of units by selling which the Co. will neither loss nor gain anything.

(ii) The sales needed to earn profit of 20% on sales.

(iii) The extra unit which should be sold to obtain the present profit it is proposed to reduce the selling price by 20%.

(iv) The selling price to be fixed to bring down its break-even point to 1,000 units under present condition.

Solution : ***Page No-164; Problem No-35***

17. From the following information prepare a cash budget for the three months commencing 01 June, 2022. The bank balance on that date is Tk. 1,10,000 :

Details	April (Tk.)	May (Tk.)	June (Tk.)	July (Tk.)	August (Tk.)
Sales	88,500	84,000	93,000	72,000	82,000
Selling overhead	3,250	4,100	3,710	3,210	3,600
Purchases	37,000	40,000	39,000	39,000	35,000
Wages	8,000	8,400	8,800	6,000	9,600
Factory overhead	5,680	5,920	2,440	5,880	6,000
Administrative overhead	2,500	2,750	2,480	2,600	2,520

Others information :

(i) A sales commission of 2% on sales is due 2 months after sales in payable in addition to the above selling overhead.

(ii) Plant purchases June 01 for Tk. 38,000, payable on delivery.

(iii) Building purchases in the month of June for Tk. 80,000, payable in two half yearly installment, the 1st in the month of July.

(iv) A dividend of Tk. 20,000 will be paid in July

(v) Sales and purchases are made 10% on cash. Credit allowed by suppliers is one month. Debtors are allowed two months credits.

(vi) Time lag-in-payments wages and all other overheads is 15 days. **Solution :** ***Page No-250; Problem No-30***

National University

[BBA (Hon's) 3rd Year; Exam-2023 (Held in-04/05/2025)]

(Department of Accounting)

Subject Code : 232507

Subject Name : Management Accounting (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Answer different questions of each group chronologically.]

Part-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is management process?

Ans : The management process is one of the methods that aid the structuring, investigation, analysis, decision making and communicate of business issues.

(b) Elaborate IFAC.

Ans : International Federation of Accountants.

(c) What is imputed cost?

Ans : A cost that is incurred by virtue of using an asset instead of investing it or undertaking an alternative is called imputed cost.

(d) What is segment reporting?

Ans : An income statement or other report in an organization in which data are divided according to product line, division, territories or similar organizational segments.

(e) What is rolling budget?

Ans : Rolling budget is a form of master budgets that simply adds a month in future as the month just ended is dropped.

(f) Mention the two non-discounting techniques of capital budgeting.

Ans : Two non-discounting techniques of capital budgeting are (i) payback period method & (ii) Average rafe of return method.

(g) What is cost indifference point?

Ans : The cost indifference point is the level of activity at which the total costs of two alternative options are equal.

(h) What is cross-subsidization?

Ans : Cross-subsidization occurs when the profits from one product, service or customer segment are used to cover the losses lower prices of another.

(i) Give two examples of static cost.

Ans : Two examples of static cost are (i) Monthly office rent & (ii) Salaries of permanent staff.

(j) What is relevant cash flow?

Ans : The relevant cash flows is future incremental cash flow arising from the decision being made. This means that the cash flows are only relevant if they are future, incremental costs.

(k) What is target costing?

Ans : Target costing implies developing product cost in terms of what the market will pay for a product with specific characteristics.

(l) What is variable overhead spending variance?

Ans : Variable overhead spending variance is the differences between the actual variable overhead cost incurred during a period and the standard cost that should have been incurred based on the actual activity of the period.

Part-B

[Answer any **five** questions.] Marks : $4 \times 5 = 20$

2. "Management accounting has a broader coverage which includes both internal and external considerations."–Explain.

[Ans : ব্যবস্থাপনা হিসাববিজ্ঞান হাজ্জাক (৩য় খণ্ড) বইয়ের Page No. 13; Ques. No. 1.27]

3. "If a product line is generating a loss, then it should be discontinued."–Do you agree? Explain.

[Ans : Page No. 25 ; Ques. No. 2.27]

4. When and why profits will be higher in case of absorption costing and variable costing?

[Ans : Page No. 76 ; Ques. No. 3.15]

5. The selling price of a product is Tk. 20 per unit and contribution margin is 25%. If Tk. 15,000 profit is made by selling 15,000 units of a product, calculate break even sales Tk.

Solution : ***Page No-146; Problem No-05***

6. The following data relates to the Horizon Manufacturing Company of a standard product during four weeks period of June 30, 2024 :

Account titles	Tk.
Raw material used	Tk. 5,000
Wages	Tk. 7,000
Machine hours worked 2,000 hours	
Machine hour rate	Tk. 1.00
Office overhead : 25% on work cost	
Selling overhead per units	Tk. 0.10
Units produced :	25,000 units
Units sold :	2,000 units @ Tk. 1 per unit

Required : (i) Cost sheet; (ii) Income statement.

Solution : ***Page No-52; Problem No-29***

7. Rangdhonu Ltd. produced 32,000 units of product during the year 2022, 27,000 units were sold @ Tk. 200 each. Cost of this product was :

Material Tk. 7,00,000. Labour (direct) Tk. 5,00,000

Factory OH :	
Fixed	Tk. 1,52,000
Variable	Tk. 1,25,000

Marketing and administrative expenses for the year total Tk. 1,10,000, all are fixed. You are required production cost per unit using variable costing method.

Solution : ***Page No-78; Problem No-02***

8. The Sagor Company is using 15,000 units of part L-5 as a component to assemble one of its products. It costs the company to produce it internally having itemized expenditures as follows :

Direct material	Tk. 1,50,000
Direct labor	Tk. 1,55,000
Variable overhead	Tk. 90,000
Fixed overhead	Tk. 3,00,000

An outside vendor has just offered to supply the part for Tk. 35 per unit. If the company stops producing this part, $\frac{1}{4}$ of the fixed overhead would be avoided.

Should the company make or buy this part?

Solution : ***Page No-194; Problem No-04***

9. Ratul Enterprise supplied you the following information about its production and sales :

Selling price Tk. 16 per unit
Variable cost Tk. 8 per unit
Production is 10,000 units
Beginning inventory was 8,000 units
Ending inventory is 6,000 units
Variable selling and administration expense Tk. 0.5 per unit
Fixed selling and administration expense Tk. 10,000
Fixed factory overhead was Tk. 40,000 for 20,000 units.

You are required to :

(i) Find net income under absorption costing
(ii) Explain the reason for the difference in net income between absorption costing and direct costing method.

Solution : ***Page No-83; Problem No-12***

Part-C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. (a) "Value of inventory under full costing is greater than variable costing"–Explain with example.

[Ans : Page No. 75; Ques. No. 3.13]

(b) Why relevant cost is an important Managerial tool? Explain with example.

[Ans : Page No. 191 ; Ques. No. 6.19]

11. (a) "Standard costing is the tools for effective control of cost." –Explain the statement.

[Ans : Page No. 321 ; Ques. No. 9.12]

(b) From the following information, calculate labour variance.

Standard cost per unit :
10 hours Tk. 90 per hour
Actual cost for producing 200 units :
2,100 hours @ Tk. 80 per hour = Tk. 1,68,000.
50 hours are lost due to abnormal losses.

Solution : ***Page No-351; Problem No-22***

12. In 2022 margin of safety, contribution margin ratio and margin of safety ratio of Korotowa Ltd. were Tk. 40,000, 25% and 20% respectively. In 2023 Meghna Ltd. wants to sales 24,000 units by increasing it sales by 20%. It is expected that in 2023, unit selling price will increase by 10%, variable cost per unit will increase by Tk. 0.20 and fixed cost will decrease by 10% Tax rate 40%.

Required :

(a) Calculate break even sales, actual sales and profit % for 2022 and 2023.
(b) Determine margin of safety ratio for 2023.
(c) Determine the target sales to earn after tax profit Tk. 14,400.

Solution : ***Page No-179; Problem No-50***

13. The below data pertain to August 2023 :

Beginning inventory	**Nill**
Units sold	5,000 units
Units produced	8,000 units
Sales per unit	Tk. 24
Direct manufacturing cost (per unit)	Tk. 10
Fixed factory overhead per unit	Tk. 3.50
Fixed factory overhead (total)	Tk. 28,000
Selling and distribution expense (all are fixed)	Tk. 10,000

Required :

(a) Prepare two income statements using direct costing and absorption costing
(b) Prepare reconciliation statement between two methods.

Solution : ***Page No-106; Problem No-30***

14. The expenses budget for the production of 10,000 units in a factory are as follows :

Particulars	**Pet unit**
Raw material	Tk. 90.00
Labour	Tk. 25.00
Variable overhead	Tk. 20.00
Fixed overhead (Tk. 1,00,000)	Tk. 10.00
Variable expense (Direct)	Tk. 5.00

Selling expenses (10% fixed)	Tk. 15.00
Distribution expense (20% fixed)	Tk. 10.00
Administrative expense (Tk. 60,000)	Tk. 6.00

You are required to prepare a flexible budget for the production of 5,000 units and 8,000 units.

Solution : ***Page No-295; Problem No-05***

15. A manufacturing company is considering an older machine which was fully depreciated for tax purposes with a new machine costing Tk. 50,000. The new machine will be depreciated over its 10 years life. It is estimated that the new machine will reduce labour costs by Tk. 10,000 per year. The Management believes that there will no change in expenses and revenues of the firm due to the machine. The company requires an after tax return on investment of 10% and tax rate 35%. The company's income statement for the current year is given for other information's:

Particulars	Tk.	Tk.
Sales		5,00,000
Less : costs :		
Material	1,50,000	
Labour	2,00,000	
Factory and administrative	42,000	
Depreciation	40,000	(4,32,000)
Net income before tax		68,000
Less : Taxes (35%)		(23,800)
Earning after taxes		44,200

Should the company buy the new machine? Assume that the company follows straight line method and same tax rate is allowed for purchasing new machine.

Solution : ***Page No-271; Problem No-53***

16. A company is considering a project with required an investment Tk. 4,00,000 and also need Tk. 50,000 as a working capital. After life time the estimated salvage value Tk. 30,000. Tax rate is 50%. The company uses straight line depreciation and proposed project has cash flow before tax (CFBT) Tk. 1,70,000 per annum

Determine the following :

(a) Pay back period

(b) Average rate of return

(c) Net present value (Assume cost of capital is 15%)

(d) Internal rate of return

Solution : ***Page No-261; Problem No-44***

17. Standard material for 100 units are given below :

Material P	45 units @ Tk. 2 each	Tk. 90
Material Q	40 units @ Tk. 4 each	Tk. 160
Material R	25 units @ Tk. 6 each	Tk. 150
	110	Tk. 400
Standard loss	(10)	
	100	

Actual production 2,000 units and actual material usages is as follows :

Material P	1,000 units @ Tk. 1.90	= Tk. 1,900
Material Q	850 units @ Tk. 4.20	= Tk. 3,570
Material R	450 units @ Tk. 6.50	= Tk. 2,925
	2,300 units	= Tk. 8,395

Calculate material cost variance, material price variance, material mix variance, material yield variance and material usage variance.

Solution : ***Page No-346; Problem No-18***

National University

[BBA Hon's 3rd Year; Exam-2024 (Held on- 07/01/2026)]

(Department of Accounting)

Subject Code : 232507

Subject Name : Management Accounting (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question must be written consecutively.]

Part-A

1. Answer any **ten** questions : $1 \times 10 = 10$

(a) Write the name of three types of costs under cost classification for predicting cost behavior.

Ans : The name of three types of costs under cost classification for predicting cost behavior are variable cost, fixed cost and semi-variable cost.

(b) What is mixed cost?

Ans : Mixed cost is a cost that contains both fixed and variable components.

(c) What is Product Lifecycle Management (PLM)?

Ans : Product life cycle management is the succession of strategies used by business management as product goes through its life cycle.

(d) In which case, there in no difference in net income under variable costing and absorption costing?

Ans : There is no difference in net income under variable costing and absorption costing when there is no change in inventory levels.

(e) **What is angle of incidence?**
Ans : In graphical representation of cost-volume-profit analysis the angle which is formed in the right side by the intersection of sales line with the total cost line is known as the angle of incidence.

(f) **Wht is relevant cost?**
Ans : Relevant cost are those expected future cost that differ among alternative course of action.

(g) **What is participatory budgeting?**
Ans : Participatory budgeting is a budgeting approach in which managers and employees at different levels actively participate in the preparation of the budget.

(h) **A project's NPV at 10% discount rate is zero (0), what will be the IRR?**
Ans : A project's NPV at 10% discount rate is zero (0), then the IRR is 10%.

(i) **What is activity based costing?**
Ans : Activity based-costing refers a costing system by identifying individual activities as the fundamental cost objects.

(j) **What is flexible budget?**
Ans : A budget which is designed to changed in accordance with the level of activity actually attained is called flexible budget.

(k) **What is labor efficiency variance?**
Ans : Labour efficiency variance is the difference between the standard hours for the actual production achieved and the hours actually worked, valued at the standard labour rate.

(l) **What is zero based budget?**
Ans : The budget which comes from zero is called zero based budget.

Part-B

[Answer any **five** questions.] Marks : 4 × 5 = 20

2. Distinguish between a master budget and a sales forecast.
[Ans : ব্যাভিকল্প হিসাববিজ্ঞান হ্যান্ডনোট (৩য় খণ্ড) বইয়ের Page No. 634 Question No. 2]

3. Briefly describe the prospect of management accounting profession in Bangladesh.
[Ans : Page No. 634 Question No. 3]

4. Show the differences between simple costing system and activity based costing system.
[Ans : Page No. 125; Ques. No. 4.07]

5. Discuss the methods of segregation of semi variable cost. ***[Ans : Page No. 635; Question No. 5]***

6. The following figures for the year ended December 31, 2022 were extracted from the records of a manufacturing company :

Inventory of finished goods (1-1-2022) (5,000 units)	45,000
Raw materials purchase	2,60,000
Direct wages	1,00,000
Sales (45,000 units)	7,00,000

Ending finished goods (10,000 units). Factory overhead 60% of direct wages. Administrative overhead Tk. 50,000, selling overhead Tk. 2 per unit.

Prepare a cost sheet for the year ended December 31, 2022. **Solution :** ***Page No-635; Question No-06***

7. ACI Ltd. produces a single product and supplies the following cost data :

Production during the year	20,000 units
Sales during the year	15,000 units
Direct material cost	Tk. 1,50,000
Direct labour cost	Tk. 1,00,000
Variable factory overhead	Tk. 60,000
Fixed factory overhead	Tk. 40,000

Required :
(a) Cost per unit and value of ending inventory under variable costing.
(b) Cost per unit and value of ending inventory under absorption costing.

Solution : ***Page No-636; Question No-07***

8. Rakib company for the last year disclosed an M/S ratio of 20% and C/M ratio of 60%. Fixed cost Tk. 60,000 Calculate : (i) BEP (Tk.) (ii) Profit (iii) CM.

Solution : ***Page No-637; Question No-08***

9. XY Ltd. is examining the profitability of investing in new project. The following information is available in respect of the project :

Investment	Tk. 20,00,000
Salvage value	Tk 1,00,000
Life of project	20 years
Annual profit of the project (before tax) is	Tk. 3,50,000

Calculate : (i) PBP (ii) ARR.

Solution : ***Page No-637; Question No-09***

Part-C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. (a) "Management Accounting is the best tools for the management to achieve higher profits and efficient operation". Discuss. ***[Ans : Page No. 638; Question No. 10 (a)]***

(b) Discuss the objectives of preparing cost sheet. ***[Ans : Page No. 638; Question No. 10 (b)]***

11. (a) What are the techniques of management accounting? ***[Ans : Page No. 7; Ques. No. 1.10]***

(b) Distinguish between financial accounting and management accounting. ***[Ans : Page No. 8; Ques. No. 1.13]***

12. The following particulars are available from the books of M Ltd. for the year ended 2020 :

Sales 1,50,000 units @ Tk. 30 per unit, Beginning finished goods 24,000 units, Ending finished goods 35,000 units. Manufacturing cost :

Variable cost per unit Tk. 12, Fixed factory overhead Tk. 1,60,000, Normal capacity 1,60,000 units, variable marketing and administrative expenses Tk. 1.50 per unit of sales. Fixed administrative expenses Tk. 3,00,000.

Required :

(a) Income statement under direct costing.

(b) Income statement under absorption costing.

(c) Reconciliation statement.

Solution : ***Page No-639; Question No-12***

13. Sakib Ltd. furnishes you the following data relating to the year 2024 :

	First half of the year	Second half of the year
Sales	Tk. 50,000	Tk. 80,000
Total cost	Tk. 40,000	Tk. 55,000

Assuming that there is no change in prices and variable cost and fixed cost are incurred equally in the two half year periods, calculate for the year 2024 :

(i) P/V ratio; (ii) Fixed cost; (iii) BEP sales; (iv) Sales required earning profit Tk. 30,000; (v) Profit when annual sales Tk. 1,00,000.

Solution : ***Page No-641; Question No-13***

14. The standard cost and actual cost of materials per unit are given below :

Standard cost :

Material A : 5 kg @ Tk. 8

Material B : 10 kg @ Tk. 6

Actual cost :

Material A : 8 kg @ Tk. 7

Material B : 8.5 kg @ Tk. 7

During a week, 400 units are produced calculate :

(i) Materials price variance

(ii) Material mix variance

(iii) Material usage variance

(iv) Material cost variance

(v) Material yield variance

Solution : ***Page No-642; Question No-14***

15. From the information given below prepare a cash budget of X Ltd. for 1st quarter 2024. Cash balance 1st January Tk. 30,000. Monthly wage and Salary Tk. 10,000, debenture interest payable in March 2024 Tk. 10,000. Other information are given below :

Details	December	January	February	March
Cash sales	80,000	1,40,000	1,50,000	1,21,000
Credit sales	1,00,000	80,000	1,40,000	1,20,000
Purchase	1,60,000	1,80,000	2,40,000	1,80,000
Other payment	16,000	20,000	22,000	20,000

Terms of purchases and sales :

(i) 10% of the purchases are paid promptly and the remaining in 30 days basis.

(ii) Credit sales are collected 50% in the month of sales are made and 50% in the following month, collection from credit sales are subject to 5% discount if payment is received during the month of sales and 2.5% discount if payment is received in the month following.

Solution : ***Page No-643; Question No-15***

16. The following are the monthly budget for the manufacturing overhead of a concern for two levels of activity :

Capacity	60%	100%
Budgeted production (units)	600	1,000
Indirect wages	Tk. 1,20,000	Tk. 2,00,000
Consumable store	90,000	1,50,000
Maintenance	1,10,000	1,50,000
Power and fuel	1,60,000	2,00,000
Depreciation	6,00,000	6,00,000
Insurance	2,00,000	2,00,000

Required :

(a) Indicate which of the items are fixed, variable and mix.

(b) Prepare a flexible budget for 70%, 80% and 90% activity level.

Solution : ***Page No-644; Question No-16***

17. A company has three product lines, all are produced in one factory. Management is deep concerned with product C which has consistently shown a net loss. The income statement is given below :

Particulars	A	B	C	Total
Sales	5,00,000	4,00,000	1,00,000	10,00,000
Less : Variable cost	2,95,000	2,80,000	75,000	6,50,000
Contribution margin	2,05,000	1,20,000	25,000	3,50,000
Fixed expenses (Salaries, depreciation etc.)	1,65,000	90,000	45,000	3,00,000
Net income (Loss)	40,000	30,000	(20,000)	50,000

Required :

(i) Whether product C will be dropped when only alternative is to drop product C or to continue with it. Assume that fixed expenses of product includes Tk. 20,000 supervision salary. Assume further that the total assets invested will not be affected by the decision.

(ii). Drop product C, keeping the supervision and using the vacant facilities to product A to satisfy its expanding demand by an additional sales of Tk.1,00,000. Which alternative is more profitable.

Solution : ***Page No-646; Question No-17***